

SCEM Final Coursework

Task II

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IMPORTANT NOTES:

- **DO NOT** change the code block names. Enter your solutions to each question into the predefined code blocks.
- **Don't forget to replace the placeholder value in code block "Setup" by your valid student ID number.**
- If you want to use any packages, add their names to the vector `required.packages` in the code block "Setup". That code block will automatically install those packages (if needed) and load them. Any other calls to `install.packages()`, `library()` or `require()` may result in zero marks for the activity where it happens.
- Make sure that the data files `df.rds` and `df_holdout.rds` are located in the same folder as this file, `Task02.Rmd`.
- Instructions for this task are provided in blue text below. Your solution should NOT be added within those text boxes, and should NOT be in blue text. **During marking, all the text within the text blocks below, with names starting with "prompt", will be removed and not visible to the marker.**

```
##  
## List of allowed packages (pre-loaded):  
##   tidyverse  
##   tidymodels  
##   recipes  
##   parsnip  
##   bonsai  
##   healthyR.ai  
##   knitr  
##   DataExplorer  
##   visdat
```

In this task you will build a full predictive pipeline for the problem described in the coursework specs. Please read each activity carefully and follow the instructions to complete your coursework.

Make sure that the data file **df.rds** is placed in the same folder as this Task02.Rmd file. The data that you can use for model development in this task will then be automatically loaded and set up for you, and stored into a dataframe variable called **df**.

1. Exploratory Data Analysis

Note: It is expected that this section should result in no more than 600-ish words (possibly less; not counting the code blocks).

Add here your exploratory data analysis. Your solution should include both your code (which markers must be able to run independently) and your rationale for each exploratory step. In general, your EDA steps should include:

- A short justification of what a given plot or statistical summary is intended to reveal about the data.
- A code block implementing your EDA step.

At the end of your EDA section, you should also include a short commentary on what your data exploration revealed about the data, and which pre-processing steps may be required based on that.

You can use as many code blocks as needed, but don't overdo it (remember - everything needs to have a proper rationale).

1.1 Data overview

The dataset is derived from a binary epitope prediction problem, where each observation is described by a large set of numerical predictive features. An exploratory data analysis provided an initial overview of potential dependencies caused by "Info_group", distribution of the target "Class", and the behavior of features "feat_" in terms of scale, variability and missingness.

Quick overview shows that some of the columns that begin with "Info_" are mainly identifiers or meta data. As these variables do not carry predictive information about the target data, discarding them could be considered. Moreover, further investigation is required as the presence of some missing values and variables with very different scales could be identified from the dataset. Data preprocessing such as imputation and standardisation can be used to ensure that the predictors are on comparable scales and that missing values do not bias the prediction model.

```
## Rows: 7,403
## Columns: 390
## $ Info_PepID          <chr> "QLJ58341.1:11", "P06477.1:4", ~
## $ Info_group          <int> 61, 357, 467, 239, 9, 55, 150, ~
## $ Info_protein_id     <chr> "QLJ58341.1", "P06477.1", "ATQ3~
## $ Info_pos            <dbl> 491.0000, 215.5000, 186.0000, 2~
## $ feat_local_Entropy  <dbl> 3.228194, 3.067177, 3.009131, 3~
## $ feat_local_MolWeight <dbl> 2037.570, 1900.578, 1920.043, 1~
## $ feat_local_AAtypes_Tiny <dbl> 0.2493827, 0.3676923, 0.3826087~
## $ feat_local_AAtypes_Small <dbl> 0.4296296, 0.6210256, 0.6057971~
## $ feat_local_AAtypes_Aliphatic <dbl> 0.24691358, 0.28871795, 0.32173~
## $ feat_local_AAtypes_Aromatic <dbl> 0.138271605, 0.122051282, 0.144~
## $ feat_local_AAtypes_NonPolar <dbl> 0.4839506, 0.5882051, 0.5072464~
## $ feat_local_AAtypes_Polar <dbl> 0.5160494, 0.4117949, 0.4927536~
## $ feat_local_AAtypes_Charged <dbl> 0.26172840, 0.21128205, 0.15652~
## $ feat_local_AAtypes_Basic <dbl> 0.07654321, 0.13641026, 0.09565~
## $ feat_local_AAtypes_Acidic <dbl> 0.18518519, 0.07487179, 0.06086~
## $ feat_local_Atoms_nC <dbl> 80.37037, 73.13077, 74.60870, 7~
## $ feat_local_Atoms_nH <dbl> 147.5556, 142.9538, 146.2174, 1~
## $ feat_local_Atoms_nN <dbl> 19.48148, 21.30769, 18.47826, 2~
## $ feat_local_Atoms_nO <dbl> 39.70370, 35.60000, 37.73913, 3~
```

## \$ feat_local_Atoms_nS	<dbl> 0.4814815, 0.3153846, 0.4347826~
## \$ feat_local_AAC_A	<dbl> 0.069135802, 0.100000000, 0.037~
## \$ feat_local_AAC_R	<dbl> 0.002469136, 0.098974359, 0.043~
## \$ feat_local_AAC_N	<dbl> 0.017283951, 0.017948718, 0.040~
## \$ feat_local_AAC_D	<dbl> 0.083950617, 0.042564103, 0.034~
## \$ feat_local_AAC_C	<dbl> 0.000000000, 0.007692308, 0.028~
## \$ feat_local_AAC_E	<dbl> 0.101234568, 0.032307692, 0.026~
## \$ feat_local_AAC_Q	<dbl> 0.096296296, 0.000000000, 0.000~
## \$ feat_local_AAC_G	<dbl> 0.039506173, 0.077435897, 0.020~
## \$ feat_local_AAC_H	<dbl> 0.000000000, 0.037435897, 0.008~
## \$ feat_local_AAC_I	<dbl> 0.06172840, 0.02153846, 0.06086~
## \$ feat_local_AAC_L	<dbl> 0.06419753, 0.09076923, 0.07536~
## \$ feat_local_AAC_K	<dbl> 0.074074074, 0.000000000, 0.043~
## \$ feat_local_AAC_M	<dbl> 0.032098765, 0.013333333, 0.000~
## \$ feat_local_AAC_F	<dbl> 0.01234568, 0.03076923, 0.07826~
## \$ feat_local_AAC_P	<dbl> 0.027160494, 0.116410256, 0.000~
## \$ feat_local_AAC_S	<dbl> 0.10617284, 0.08256410, 0.23768~
## \$ feat_local_AAC_T	<dbl> 0.03456790, 0.10000000, 0.05797~
## \$ feat_local_AAC_W	<dbl> 0.10370370, 0.03076923, 0.00000~
## \$ feat_local_AAC_Y	<dbl> 0.022222222, 0.023076923, 0.057~
## \$ feat_local_AAC_V	<dbl> 0.05185185, 0.07641026, 0.14782~
## \$ feat_local_CTDC_hydrophobicity.Group1	<dbl> 0.3753086, 0.1917949, 0.1884058~
## \$ feat_local_CTDC_hydrophobicity.Group2	<dbl> 0.2987654, 0.5369231, 0.4202899~
## \$ feat_local_CTDC_hydrophobicity.Group3	<dbl> 0.32592593, 0.27128205, 0.39130~
## \$ feat_local_CTDC_normwaalsvolume.Group1	<dbl> 0.3604938, 0.5266667, 0.4173913~
## \$ feat_local_CTDC_normwaalsvolume.Group2	<dbl> 0.3925926, 0.2389744, 0.3507246~
## \$ feat_local_CTDC_normwaalsvolume.Group3	<dbl> 0.24691358, 0.23435897, 0.23188~
## \$ feat_local_CTDC_polarity.Group1	<dbl> 0.34814815, 0.29435897, 0.44927~
## \$ feat_local_CTDC_polarity.Group2	<dbl> 0.2765432, 0.4764103, 0.3536232~
## \$ feat_local_CTDC_polarity.Group3	<dbl> 0.3753086, 0.2292308, 0.1971014~
## \$ feat_local_CTDC_polarizability.Group1	<dbl> 0.33333333, 0.40256410, 0.38840~
## \$ feat_local_CTDC_polarizability.Group2	<dbl> 0.4197531, 0.3630769, 0.3797101~
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## \$ feat_local_CTDC_charge.Group1	<dbl> 0.07654321, 0.09897436, 0.08695~
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## \$ feat_local_CTDC_charge.Group3	<dbl> 0.18518519, 0.07487179, 0.06086~
## \$ feat_local_CTDC_secondarystruct.Group1	<dbl> 0.4395062, 0.3728205, 0.2347826~
## \$ feat_local_CTDC_secondarystruct.Group2	<dbl> 0.28641975, 0.29025641, 0.43188~
## \$ feat_local_CTDC_secondarystruct.Group3	<dbl> 0.27407407, 0.33692308, 0.33333~
## \$ feat_local_CTDC_solventaccess.Group1	<dbl> 0.40246914, 0.43538462, 0.44927~
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## \$ feat_local_CTDC_solventaccess.Group3	<dbl> 0.22222222, 0.37282051, 0.36231~
## \$ feat_local_CTDD_prop1.G1.residue0	<dbl> 14.81481, 27.58974, 29.56522, 2~
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## \$ feat_local_CTDD_prop1.G1.residue50	<dbl> 39.01235, 33.48718, 31.88406, 3~
## \$ feat_local_CTDD_prop1.G1.residue75	<dbl> 56.79012, 51.17949, 57.39130, 5~
## \$ feat_local_CTDD_prop1.G1.residue100	<dbl> 86.17284, 75.23077, 84.63768, 8~
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## \$ feat_local_CTDD_prop1.G2.residue100	<dbl> 86.91358, 94.71795, 93.62319, 8~
## \$ feat_local_CTDD_prop1.G3.residue0	<dbl> 16.29630, 24.05128, 14.78261, 1~
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## \$ feat_local_CTDD_prop1.G3.residue50	<dbl> 38.51852, 36.92308, 43.76812, 4~
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## \$ feat_local_CTDD_prop2.G1.residue0	<dbl> 15.061728, 12.615385, 13.623188~
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## \$ feat_local_CTDD_prop6.G1.residue75	<dbl> 59.50617, 58.10256, 56.81159, 5~
## \$ feat_local_CTDD_prop6.G1.residue100	<dbl> 87.90123, 87.79487, 84.34783, 8~
## \$ feat_local_CTDD_prop6.G2.residue0	<dbl> 19.259259, 23.128205, 12.753623~
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## \$ feat_local_CTDD_prop6.G2.residue100	<dbl> 85.43210, 84.76923, 91.88406, 9~
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## \$ feat_local_CTDD_prop6.G3.residue50	<dbl> 38.76543, 37.53846, 42.60870, 3~
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## \$ feat_local_CTDD_prop6.G3.residue100	<dbl> 87.40741, 88.15385, 93.91304, 7~
## \$ feat_local_CTDD_prop7.G1.residue0	<dbl> 17.037037, 15.641026, 14.782609~
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## \$ feat_local_CTDD_prop7.G2.residue0	<dbl> 14.81481, 27.58974, 29.56522, 2~
## \$ feat_local_CTDD_prop7.G2.residue25	<dbl> 15.80247, 27.58974, 29.56522, 2~
## \$ feat_local_CTDD_prop7.G2.residue50	<dbl> 39.01235, 33.48718, 31.88406, 3~
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## \$ feat_local_CTDD_prop7.G3.residue0	<dbl> 28.395062, 19.692308, 11.594203~
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## \$ feat_local_CTDI_prop1.Tr1221	<dbl> 0.15873016, 0.24340659, 0.16459~
## \$ feat_local_CTDI_prop1.Tr1331	<dbl> 0.22222222, 0.07857143, 0.21428~
## \$ feat_local_CTDI_prop1.Tr2332	<dbl> 0.27248677, 0.28736264, 0.47515~
## \$ feat_local_CTDI_prop2.Tr1221	<dbl> 0.41269841, 0.20879121, 0.36956~
## \$ feat_local_CTDI_prop2.Tr1331	<dbl> 0.18253968, 0.26318681, 0.25776~
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## \$ feat_local_CTDI_prop5.Tr2332	<dbl> 0.26190476, 0.13406593, 0.12422~

```

## $ feat_local_CTDt_prop6.Tr1221 <dbl> 0.20105820, 0.16208791, 0.22670~
## $ feat_local_CTDt_prop6.Tr1331 <dbl> 0.19841270, 0.25824176, 0.22981~
## $ feat_local_CTDt_prop6.Tr2332 <dbl> 0.16137566, 0.17472527, 0.33229~
## $ feat_local_CTDt_prop7.Tr1221 <dbl> 0.22222222, 0.17857143, 0.29192~
## $ feat_local_CTDt_prop7.Tr1331 <dbl> 0.24603175, 0.29230769, 0.43478~
## $ feat_local_CTDt_prop7.Tr2332 <dbl> 0.15873016, 0.14340659, 0.08695~
## $ feat_local_BLOSUM_scl1.lag1 <dbl> 0.003684638, 0.003980041, -0.01~
## $ feat_local_BLOSUM_scl2.lag1 <dbl> -1.203654e-02, 1.824008e-03, 1.~
## $ feat_local_BLOSUM_scl3.lag1 <dbl> 4.943067e-03, -6.319988e-03, -7~
## $ feat_local_BLOSUM_scl4.lag1 <dbl> -0.0055321759, 0.0029844950, -0~
## $ feat_local_BLOSUM_scl5.lag1 <dbl> 0.0055066792, 0.0066485645, 0.0~
## $ feat_local_BLOSUM_scl1.lag2 <dbl> -0.0042972800, 0.0035929042, -0~
## $ feat_local_BLOSUM_scl2.lag2 <dbl> -0.0048748138, 0.0040629716, 0.~
## $ feat_local_BLOSUM_scl3.lag2 <dbl> -0.0027967653, -0.0004792421, --
## $ feat_local_BLOSUM_scl4.lag2 <dbl> 4.117121e-03, 5.906132e-04, -2.~
## $ feat_local_BLOSUM_scl5.lag2 <dbl> 7.507905e-05, 6.917388e-03, 8.3~
## $ feat_local_BLOSUM_scl1.lag3 <dbl> -0.023084816, -0.004014740, 0.0~
## $ feat_local_BLOSUM_scl2.lag3 <dbl> -0.0012875237, 0.0033785586, 0.~
## $ feat_local_BLOSUM_scl3.lag3 <dbl> 1.487295e-02, 1.061646e-03, 9.4~
## $ feat_local_BLOSUM_scl4.lag3 <dbl> -0.0147160343, -0.0004660096, 0~
## $ feat_local_BLOSUM_scl5.lag3 <dbl> -6.203827e-03, 8.612181e-03, 7.~
## $ feat_local_BLOSUM_scl1.2.lag1 <dbl> 0.0172858000, -0.0025875618, 0.~
## $ feat_local_BLOSUM_scl1.3.lag1 <dbl> 0.0002659425, -0.0008712273, -0~
## $ feat_local_BLOSUM_scl1.4.lag1 <dbl> -1.929952e-02, 1.318220e-03, -4~
## $ feat_local_BLOSUM_scl1.5.lag1 <dbl> 0.003891764, -0.004557193, -0.0~
## $ feat_local_BLOSUM_scl2.3.lag1 <dbl> -0.004890027, -0.003243064, 0.0~
## $ feat_local_BLOSUM_scl2.4.lag1 <dbl> -0.002856274, 0.008421043, 0.00~
## $ feat_local_BLOSUM_scl2.5.lag1 <dbl> 0.0089652997, -0.0025999301, -0~
## $ feat_local_BLOSUM_scl3.4.lag1 <dbl> -0.0095976926, 0.0013547380, -0~
## $ feat_local_BLOSUM_scl3.5.lag1 <dbl> 8.199971e-03, -3.548445e-03, 3.~
## $ feat_local_BLOSUM_scl4.5.lag1 <dbl> 0.0043954306, -0.0047854664, 0.~
## $ feat_local_BLOSUM_scl2.1.lag1 <dbl> -0.0086386333, -0.0023379868, --
## $ feat_local_BLOSUM_scl3.1.lag1 <dbl> -0.0129532595, -0.0012825429, 0~
## $ feat_local_BLOSUM_scl4.1.lag1 <dbl> -0.0119623815, -0.0017260687, 0~
## $ feat_local_BLOSUM_scl5.1.lag1 <dbl> 3.317173e-03, -4.371338e-03, -5~
## $ feat_local_BLOSUM_scl3.2.lag1 <dbl> -2.348305e-03, -8.423785e-03, --
## $ feat_local_BLOSUM_scl4.2.lag1 <dbl> -3.240199e-03, -3.825806e-03, --
## $ feat_local_BLOSUM_scl5.2.lag1 <dbl> -0.0028045840, 0.0094125589, -0~
## $ feat_local_BLOSUM_scl4.3.lag1 <dbl> -1.421242e-03, -1.155158e-03, 1~
## $ feat_local_BLOSUM_scl5.3.lag1 <dbl> 0.0024664097, 0.0071516756, -0.~
## $ feat_local_BLOSUM_scl5.4.lag1 <dbl> 5.410298e-03, -1.962690e-03, -9~
## $ feat_local_BLOSUM_scl1.2.lag2 <dbl> 0.0019135751, -0.0006179446, -0~
## $ feat_local_BLOSUM_scl1.3.lag2 <dbl> 1.137087e-02, 6.195172e-04, NA,~
## $ feat_local_BLOSUM_scl1.4.lag2 <dbl> -0.0036736464, 0.0041037056, 0.~
## $ feat_local_BLOSUM_scl1.5.lag2 <dbl> -2.141694e-03, -1.413421e-02, 3~
## $ feat_local_BLOSUM_scl2.3.lag2 <dbl> 0.0022685777, -0.0018307187, -0~
## $ feat_local_BLOSUM_scl2.4.lag2 <dbl> -9.059374e-03, -2.877160e-03, --
## $ feat_local_BLOSUM_scl2.5.lag2 <dbl> 0.0076581654, 0.0069236750, -0.~
## $ feat_local_BLOSUM_scl3.4.lag2 <dbl> 0.0142706345, -0.0024573887, 0.~
## $ feat_local_BLOSUM_scl3.5.lag2 <dbl> 0.0070931168, -0.0014021205, -0~
## $ feat_local_BLOSUM_scl4.5.lag2 <dbl> -0.0009829622, -0.0032373617, 0~
## $ feat_local_BLOSUM_scl2.1.lag2 <dbl> -0.002331536, 0.001547781, 0.02~
## $ feat_local_BLOSUM_scl3.1.lag2 <dbl> -0.0010587649, 0.0013550527, -0~
## $ feat_local_BLOSUM_scl4.1.lag2 <dbl> 0.0142682973, 0.0039252065, 0.0~

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## \$ feat_local_BLOSUM_scl5.1.lag2	<dbl> -0.0082975411, -0.0047052494, --
## \$ feat_local_BLOSUM_scl3.2.lag2	<dbl> -0.0126083336, -0.0021740243, 0~
## \$ feat_local_BLOSUM_scl4.2.lag2	<dbl> 3.890944e-03, -5.847375e-03, -1~
## \$ feat_local_BLOSUM_scl5.2.lag2	<dbl> 1.880984e-03, 7.038098e-03, -3~
## \$ feat_local_BLOSUM_scl4.3.lag2	<dbl> -8.357435e-03, 5.584982e-03, -9~
## \$ feat_local_BLOSUM_scl5.3.lag2	<dbl> 0.010072687, -0.004735826, 0.00~
## \$ feat_local_BLOSUM_scl5.4.lag2	<dbl> 0.0079128883, 0.0004062145, 0.0~
## \$ feat_local_BLOSUM_scl1.2.lag3	<dbl> -0.0006500277, -0.0007205506, 0~
## \$ feat_local_BLOSUM_scl1.3.lag3	<dbl> 0.0126226295, -0.0034479891, 0.~
## \$ feat_local_BLOSUM_scl1.4.lag3	<dbl> -0.0128326683, 0.0063265207, -0~
## \$ feat_local_BLOSUM_scl1.5.lag3	<dbl> 0.0103049264, -0.0028924252, -0~
## \$ feat_local_BLOSUM_scl2.3.lag3	<dbl> 0.0121396487, 0.0057245153, -0.~
## \$ feat_local_BLOSUM_scl2.4.lag3	<dbl> 0.0191629498, 0.0034958263, -0.~
## \$ feat_local_BLOSUM_scl2.5.lag3	<dbl> 0.0051215845, 0.0045394189, -0.~
## \$ feat_local_BLOSUM_scl3.4.lag3	<dbl> 0.0149367213, 0.0024349553, 0.0~
## \$ feat_local_BLOSUM_scl3.5.lag3	<dbl> 0.002725885, 0.001071408, 0.006~
## \$ feat_local_BLOSUM_scl4.5.lag3	<dbl> 5.294455e-03, -2.141816e-03, 9.~
## \$ feat_local_BLOSUM_scl2.1.lag3	<dbl> 0.0160167759, -0.0011793674, 0.~
## \$ feat_local_BLOSUM_scl3.1.lag3	<dbl> -0.0030696539, -0.0032998591, 0~
## \$ feat_local_BLOSUM_scl4.1.lag3	<dbl> -1.125881e-02, 4.194798e-03, -1~
## \$ feat_local_BLOSUM_scl5.1.lag3	<dbl> -1.703809e-02, -4.250882e-03, 5~
## \$ feat_local_BLOSUM_scl3.2.lag3	<dbl> 5.925101e-03, 2.136871e-04, -7.~
## \$ feat_local_BLOSUM_scl4.2.lag3	<dbl> 1.401034e-03, -3.553220e-03, 2.~
## \$ feat_local_BLOSUM_scl5.2.lag3	<dbl> 0.0035584512, 0.0016468059, -0.~
## \$ feat_local_BLOSUM_scl4.3.lag3	<dbl> -0.012640197, -0.002909482, 0.0~
## \$ feat_local_BLOSUM_scl5.3.lag3	<dbl> -0.0070285915, -0.0016713254, 0~
## \$ feat_local_BLOSUM_scl5.4.lag3	<dbl> -7.298812e-03, -6.646995e-03, 6~
## \$ feat_local_SOCN_Schneider.lag1	<dbl> 4.945126, 4.892777, 5.067505, 4~
## \$ feat_local_SOCN_Schneider.lag2	<dbl> 5.026021, 4.331752, 4.451679, 4~
## \$ feat_local_SOCN_Schneider.lag3	<dbl> 5.559105, 4.683561, 2.807200, 3~
## \$ feat_local_SOCN_Grantham.lag1	<dbl> 158957.37, 125615.98, 195782.30~
## \$ feat_local_SOCN_Grantham.lag2	<dbl> 142720.26, 115691.28, 168475.96~
## \$ feat_local_SOCN_Grantham.lag3	<dbl> 168330.22, 122701.92, 105391.30~
## \$ feat_local_QSO_Schneider.Xr.A	<dbl> 0.4068185, 0.6333393, 0.2769243~
## \$ feat_local_QSO_Schneider.Xr.R	<dbl> 0.01699315, 0.58883979, 0.27337~
## \$ feat_local_QSO_Schneider.Xr.N	<dbl> 0.10503747, 0.11492801, 0.31916~
## \$ feat_local_QSO_Schneider.Xr.D	<dbl> 0.49038115, 0.26837462, 0.25692~
## \$ feat_local_QSO_Schneider.Xr.C	<dbl> 0.00000000, 0.04701391, 0.17697~
## \$ feat_local_QSO_Schneider.Xr.E	<dbl> 0.56874022, 0.20105395, 0.15715~
## \$ feat_local_QSO_Schneider.Xr.Q	<dbl> 0.58117296, 0.00000000, 0.00000~
## \$ feat_local_QSO_Schneider.Xr.G	<dbl> 0.28482583, 0.48654403, 0.15985~
## \$ feat_local_QSO_Schneider.Xr.H	<dbl> 0.00000000, 0.2325608, 0.0519883~
## \$ feat_local_QSO_Schneider.Xr.I	<dbl> 0.3449793, 0.1423855, 0.3732410~
## \$ feat_local_QSO_Schneider.Xr.L	<dbl> 0.40706712, 0.55601621, 0.53896~
## \$ feat_local_QSO_Schneider.Xr.K	<dbl> 0.40983533, 0.00000000, 0.28506~
## \$ feat_local_QSO_Schneider.Xr.M	<dbl> 0.18197158, 0.08182511, 0.00000~
## \$ feat_local_QSO_Schneider.Xr.F	<dbl> 0.07618363, 0.20207220, 0.53741~
## \$ feat_local_QSO_Schneider.Xr.P	<dbl> 0.17799901, 0.75225632, 0.00000~
## \$ feat_local_QSO_Schneider.Xr.S	<dbl> 0.6544657, 0.5583834, 1.6155587~
## \$ feat_local_QSO_Schneider.Xr.T	<dbl> 0.2193097, 0.6456119, 0.3704672~
## \$ feat_local_QSO_Schneider.Xr.W	<dbl> 0.61178651, 0.21286047, 0.00000~
## \$ feat_local_QSO_Schneider.Xr.Y	<dbl> 0.13178833, 0.14675778, 0.35411~
## \$ feat_local_QSO_Schneider.Xr.V	<dbl> 0.3024246, 0.4974616, 1.0613514~
## \$ feat_local_QSO_Grantham.Xr.A	<dbl> 2.209456e-05, 4.207406e-05, 1.3~

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## $ feat_local_QSO_Grantham.Xr.R <dbl> 8.535138e-07, 4.497040e-05, 1.3~
## $ feat_local_QSO_Grantham.Xr.N <dbl> 5.645638e-06, 8.228121e-06, 1.3~
## $ feat_local_QSO_Grantham.Xr.D <dbl> 2.634230e-05, 1.937356e-05, 1.1~
## $ feat_local_QSO_Grantham.Xr.C <dbl> 0.000000e+00, 2.175651e-06, 8.4~
## $ feat_local_QSO_Grantham.Xr.E <dbl> 3.208163e-05, 1.310697e-05, 7.4~
## $ feat_local_QSO_Grantham.Xr.Q <dbl> 3.076415e-05, 0.000000e+00, 0.0~
## $ feat_local_QSO_Grantham.Xr.G <dbl> 1.422036e-05, 3.219698e-05, 6.7~
## $ feat_local_QSO_Grantham.Xr.H <dbl> 0.000000e+00, 1.871565e-05, 2.4~
## $ feat_local_QSO_Grantham.Xr.I <dbl> 1.937892e-05, 8.735137e-06, 1.8~
## $ feat_local_QSO_Grantham.Xr.L <dbl> 2.091181e-05, 3.504486e-05, 2.5~
## $ feat_local_QSO_Grantham.Xr.K <dbl> 2.296417e-05, 0.000000e+00, 1.4~
## $ feat_local_QSO_Grantham.Xr.M <dbl> 1.028293e-05, 5.242638e-06, 0.0~
## $ feat_local_QSO_Grantham.Xr.F <dbl> 4.028689e-06, 1.251843e-05, 2.5~
## $ feat_local_QSO_Grantham.Xr.P <dbl> 9.079414e-06, 6.108662e-05, 0.0~
## $ feat_local_QSO_Grantham.Xr.S <dbl> 3.491200e-05, 3.573755e-05, 7.7~
## $ feat_local_QSO_Grantham.Xr.T <dbl> 1.127577e-05, 4.505534e-05, 1.7~
## $ feat_local_QSO_Grantham.Xr.W <dbl> 3.289960e-05, 1.155691e-05, 0.0~
## $ feat_local_QSO_Grantham.Xr.Y <dbl> 7.263434e-06, 8.798748e-06, 1.7~
## $ feat_local_QSO_Grantham.Xr.V <dbl> 1.652393e-05, 3.135096e-05, 5.0~
## $ feat_local_QSO_Schneider.Xd.1 <dbl> 0.1936831, 0.2024602, 0.2250769~
## $ feat_local_QSO_Schneider.Xd.2 <dbl> 0.1933301, 0.1790642, 0.1956687~
## $ feat_local_QSO_Schneider.Xd.3 <dbl> 0.2148681, 0.1939232, 0.1253515~
## $ feat_local_QSO_Grantham.Xd.1 <dbl> 0.3387481, 0.3493563, 0.4244320~
## $ feat_local_QSO_Grantham.Xd.2 <dbl> 0.3028371, 0.3182427, 0.3577278~
## $ feat_local_QSO_Grantham.Xd.3 <dbl> 0.3583934, 0.3323719, 0.2178186~
## $ feat_local_ScalesGap_scl1.lag1 <dbl> -5.2459097, 14.9283511, -52.661~
## $ feat_local_ScalesGap_scl2.lag1 <dbl> 12.90724995, 6.36285339, -3.372~
## $ feat_local_ScalesGap_scl3.lag1 <dbl> 21.8371382, 3.3165226, -1.38252~
## $ feat_local_ScalesGap_scl4.lag1 <dbl> -1.6831933, 9.9766005, 5.730581~
## $ feat_local_ScalesGap_scl5.lag1 <dbl> 4.6040900, 1.9492035, -0.293209~
## $ feat_local_ScalesGap_scl1.lag2 <dbl> -8.071810, 17.117671, -12.25508~
## $ feat_local_ScalesGap_scl2.lag2 <dbl> 6.4082425, 6.4223673, 2.4159722~
## $ feat_local_ScalesGap_scl3.lag2 <dbl> 12.02255419, 10.47631952, -2.05~
## $ feat_local_ScalesGap_scl4.lag2 <dbl> -5.99235366, 0.63368610, 6.2557~
## $ feat_local_ScalesGap_scl5.lag2 <dbl> 2.18268562, 0.29881374, -0.2537~
## $ feat_local_ScalesGap_scl1.lag3 <dbl> -70.004854, -13.018974, 49.1382~
## $ feat_local_ScalesGap_scl2.lag3 <dbl> 3.3743010, 5.7030687, 9.1903607~
## $ feat_local_ScalesGap_scl3.lag3 <dbl> 7.475338, -3.507207, 17.795020,~
## $ feat_local_ScalesGap_scl4.lag3 <dbl> 4.4838596, -0.4759647, 10.19220~
## $ feat_local_ScalesGap_scl5.lag3 <dbl> 1.1270369, 4.5344664, 2.4514745~
## $ feat_local_ScalesGap_scl1.2.lag1 <dbl> -3.2230075, -6.0858614, 13.6016~
## $ feat_local_ScalesGap_scl1.3.lag1 <dbl> 13.8758070, 1.3788726, 19.27253~
## $ feat_local_ScalesGap_scl1.4.lag1 <dbl> 20.97878324, 1.59716073, 18.161~
## $ feat_local_ScalesGap_scl1.5.lag1 <dbl> -13.899488, 10.239151, 10.72501~
## $ feat_local_ScalesGap_scl2.3.lag1 <dbl> -9.3753154, 6.4204741, 1.047571~
## $ feat_local_ScalesGap_scl2.4.lag1 <dbl> -0.07006430, 4.81937759, -0.052~
## $ feat_local_ScalesGap_scl2.5.lag1 <dbl> -0.7781202, -3.6126262, 0.13629~
## $ feat_local_ScalesGap_scl3.4.lag1 <dbl> -8.3860936, 0.8810462, 9.053997~
## $ feat_local_ScalesGap_scl3.5.lag1 <dbl> 5.11866203, -7.77375544, 1.6183~
## $ feat_local_ScalesGap_scl4.5.lag1 <dbl> -1.943837, 4.854601, 6.337121, ~
## $ feat_local_ScalesGap_scl2.1.lag1 <dbl> -3.32395788, -2.77279480, 28.64~
## $ feat_local_ScalesGap_scl3.1.lag1 <dbl> 15.500083, -6.300305, -17.95262~
## $ feat_local_ScalesGap_scl4.1.lag1 <dbl> -31.1567210, 14.8107423, -19.75~
## $ feat_local_ScalesGap_scl5.1.lag1 <dbl> -3.559698, 6.938563, 5.291828, ~

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## $ feat_local_ScalesGap_scl3.2.lag1 <dbl> -11.8726974, 8.8628889, 14.6800~
## $ feat_local_ScalesGap_scl4.2.lag1 <dbl> 13.9083402, 2.1639536, 10.41966~
## $ feat_local_ScalesGap_scl5.2.lag1 <dbl> -1.97884365, -4.62789504, 3.721~
## $ feat_local_ScalesGap_scl4.3.lag1 <dbl> -2.0773901, -5.8220895, -0.3158~
## $ feat_local_ScalesGap_scl5.3.lag1 <dbl> 7.32546958, -4.32068426, 0.7114~
## $ feat_local_ScalesGap_scl5.4.lag1 <dbl> 3.4097081, -0.8250872, 3.336403~
## $ feat_local_ScalesGap_scl1.2.lag2 <dbl> 12.0097415, -5.7271993, 16.7714~
## $ feat_local_ScalesGap_scl1.3.lag2 <dbl> -22.4620267, -12.0058735, -16.0~
## $ feat_local_ScalesGap_scl1.4.lag2 <dbl> 6.9553900, 0.9960779, 2.2091170~
## $ feat_local_ScalesGap_scl1.5.lag2 <dbl> -3.2770582, 13.7678463, 0.53121~
## $ feat_local_ScalesGap_scl2.3.lag2 <dbl> 0.5867838, 8.6006144, 13.328311~
## $ feat_local_ScalesGap_scl2.4.lag2 <dbl> 1.5068860, 0.8065760, 7.6717034~
## $ feat_local_ScalesGap_scl2.5.lag2 <dbl> -1.7499845, -5.9887536, 5.94713~
## $ feat_local_ScalesGap_scl3.4.lag2 <dbl> -2.523918, 7.051923, 6.158420, ~
## $ feat_local_ScalesGap_scl3.5.lag2 <dbl> 1.52688448, -1.76521269, -3.676~
## $ feat_local_ScalesGap_scl4.5.lag2 <dbl> -4.780673, 1.757202, 2.392995, ~
## $ feat_local_ScalesGap_scl2.1.lag2 <dbl> 6.4972606, -4.1172460, -6.87113~
## $ feat_local_ScalesGap_scl3.1.lag2 <dbl> -4.990632, -10.209029, 35.48380~
## $ feat_local_ScalesGap_scl4.1.lag2 <dbl> 4.699631, 16.678571, 20.914717,~
## $ feat_local_ScalesGap_scl5.1.lag2 <dbl> 3.606093, 10.119685, 14.165997,~
## $ feat_local_ScalesGap_scl3.2.lag2 <dbl> -4.4276732, 9.0594263, -2.83677~
## $ feat_local_ScalesGap_scl4.2.lag2 <dbl> 7.94499671, 1.62153116, -3.2811~
## $ feat_local_ScalesGap_scl5.2.lag2 <dbl> -6.24554693, -2.70019819, -0.95~
## $ feat_local_ScalesGap_scl4.3.lag2 <dbl> -9.4956436, -5.3094418, 11.6634~
## $ feat_local_ScalesGap_scl5.3.lag2 <dbl> 13.4488811, -10.1887254, 3.1485~
## $ feat_local_ScalesGap_scl5.4.lag2 <dbl> -0.5133840, -2.5924026, 1.44735~
## $ feat_local_ScalesGap_scl1.2.lag3 <dbl> 24.0584619, -1.3488203, -13.657~
## $ feat_local_ScalesGap_scl1.3.lag3 <dbl> -16.554448, 3.881304, 1.673103,~
## $ feat_local_ScalesGap_scl1.4.lag3 <dbl> -1.292016, -1.975828, 4.428569,~
## $ feat_local_ScalesGap_scl1.5.lag3 <dbl> -12.0129574, 11.3621587, 1.0089~
## $ feat_local_ScalesGap_scl2.3.lag3 <dbl> 2.5320080, 8.5586361, 0.8034207~
## $ feat_local_ScalesGap_scl2.4.lag3 <dbl> 3.6879269, 2.8639425, 4.1755170~
## $ feat_local_ScalesGap_scl2.5.lag3 <dbl> -1.6127996, -5.4018468, 1.76701~
## $ feat_local_ScalesGap_scl3.4.lag3 <dbl> -3.126425, 4.612992, 5.952532, ~
## $ feat_local_ScalesGap_scl3.5.lag3 <dbl> 2.7847721, -8.2288655, 8.035422~
## $ feat_local_ScalesGap_scl4.5.lag3 <dbl> -12.3737047, 2.0133137, 2.53859~
## $ feat_local_ScalesGap_scl2.1.lag3 <dbl> 19.6815346, -0.4931568, -6.2328~
## $ feat_local_ScalesGap_scl3.1.lag3 <dbl> 2.734929, -2.113433, -9.286179,~
## $ feat_local_ScalesGap_scl4.1.lag3 <dbl> 23.5057182, 0.8852892, 5.524603~
## $ feat_local_ScalesGap_scl5.1.lag3 <dbl> 12.3834033, 12.1874735, -5.7379~
## $ feat_local_ScalesGap_scl3.2.lag3 <dbl> -6.13068528, 10.40212787, 0.843~
## $ feat_local_ScalesGap_scl4.2.lag3 <dbl> 1.2296723, 3.6942596, 1.0256206~
## $ feat_local_ScalesGap_scl5.2.lag3 <dbl> -6.2099220, -5.4178486, 3.64938~
## $ feat_local_ScalesGap_scl4.3.lag3 <dbl> -5.8852335, -0.9362477, 16.8899~
## $ feat_local_ScalesGap_scl5.3.lag3 <dbl> 0.5706571, -8.2695248, 5.476243~
## $ feat_local_ScalesGap_scl5.4.lag3 <dbl> -1.41348884, 0.08840519, 4.2451~
## $ Class <dbl> 1, -1, -1, -1, -1, 1, -1, -1, --

```

The dataset is high-dimensional with 7403 rows and 390 columns including 2 discrete and 388 continuous variables. Since most of the features are numerical, no categorical encoding has to be done. Instead, preprocessing can focus on handling missing values, standardising feature scales, and applying dimensionality reduction.

```
## [ ,1]
```

```
## rows                7403
## columns              390
## discrete_columns      2
## continuous_columns   388
## all_missing_columns   0
## total_missing_values  499
## complete_rows        6918
## total_observations    2887170
## memory_usage         23837064
```

1.2 Class distribution and grouping structure

Analysis of distribution of the target variable “Class” and summary of the grouping variable “Info_group” were conducted to assess the target distribution proportion and existence of dependencies within the dataset.

```
## # A tibble: 2 x 3
##   Class Count Distribution_Proportions
##   <dbl> <int> <chr>
## 1    -1  4366 59%
## 2     1  3037 41%
```

```
##   Info_group      n
##   Min.      : 2.0   Min.      : 1.00
##   1st Qu.:148.5   1st Qu.: 1.00
##   Median :303.0   Median : 3.00
##   Mean    :311.8   Mean    : 13.53
##   3rd Qu.:471.5   3rd Qu.: 7.00
##   Max.    :660.0   Max.    :632.00
```

The distribution of the variable “Class” seems fairly balanced, showing a split of 59% for the negative and 41% for the positive class. In contrast, the summary of “Info_group” provides clear evidence of dependencies within the dataset, with the number of observations ranging from 1 to 632, with an average of approximately 13.53 observations per group. Therefore, the assumption of independence among the observations is likely to be violated if each observation is treated independently. Hence, to avoid data leakage, splitting, preprocessing and modelling should be done at the group level so that all observations from the same group are assigned either to the training or to the evaluation set.

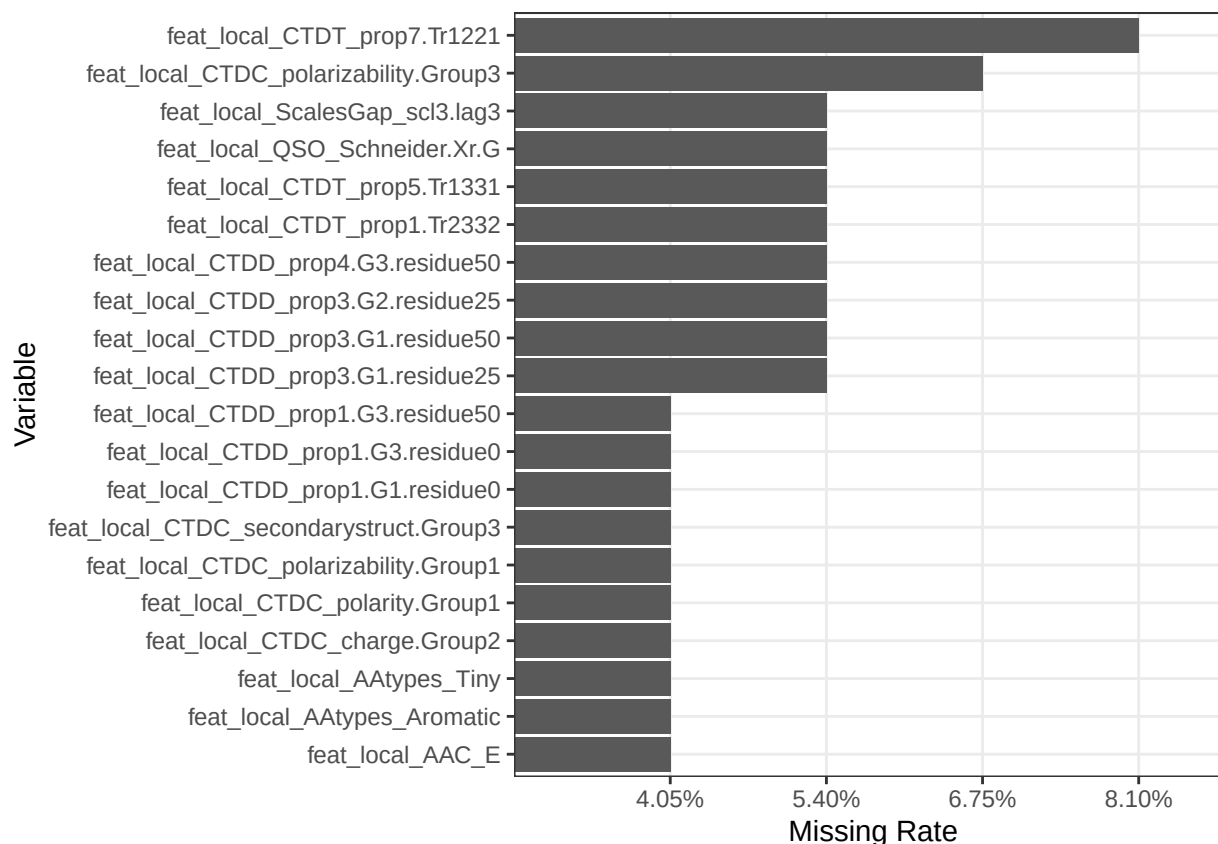
1.3 Missing values

Analysis of the missing values were conducted to assess the amount and pattern of missing values in the dataset.

```
## [1] "0.02%"

## # A tibble: 390 x 2
##   Variable      NA_ratio
##   <chr>         <chr>
## 1 Info_PepID    0.00%
## 2 Info_group    0.00%
## 3 Info_protein_id 1.35%
## 4 Info_pos      1.35%
```

```
## 5 feat_local_Entropy      2.70%
## 6 feat_local_MolWeight    2.70%
## 7 feat_local_AAtypes_Tiny  4.05%
## 8 feat_local_AAtypes_Small 0.00%
## 9 feat_local_AAtypes_Aliphatic 0.00%
## 10 feat_local_AAtypes_Aromatic 4.05%
## # i 380 more rows
```



```
##      Info_PepID      Info_pos Info_protein_id      Info_group      Class
##      0              1          1              0              0
## # A tibble: 2 x 390
##      Info_PepID      Info_group Info_protein_id Info_pos feat_local_Entropy
##      <chr>          <int> <chr>          <dbl>      <dbl>
## 1 AAF59945.1:2      90 AAF59945.1      NA        3.15
## 2 AAG22496.1:10     159 <NA>          166.      3.08
## # i 385 more variables: feat_local_MolWeight <dbl>,
## #   feat_local_AAtypes_Tiny <dbl>, feat_local_AAtypes_Small <dbl>,
## #   feat_local_AAtypes_Aliphatic <dbl>, feat_local_AAtypes_Aromatic <dbl>,
## #   feat_local_AAtypes_NonPolar <dbl>, feat_local_AAtypes_Polar <dbl>,
## #   feat_local_AAtypes_Charged <dbl>, feat_local_AAtypes_Basic <dbl>,
## #   feat_local_AAtypes_Acidic <dbl>, feat_local_Atoms_nC <dbl>,
## #   feat_local_Atoms_nH <dbl>, feat_local_Atoms_nN <dbl>, ...
```

The overall proportion of missing values ($\approx 0.02\%$) is extremely low with 499 missing values out of 2,887,170 total entries. Each variable's missing rates tend to be very low and are mostly found in a subset of feature

variables. This suggests that a simple imputation strategy, such as median imputation for numerical features, should be sufficient to handle the few missing values, without needing to drop them. Furthermore, no concerning missing values were found in the target and non-feature variables. Therefore, since missingness is limited to a few feature variables only, the next important aspect to examine is the distribution and scale of the features for feature scaling and dimensionality reduction.

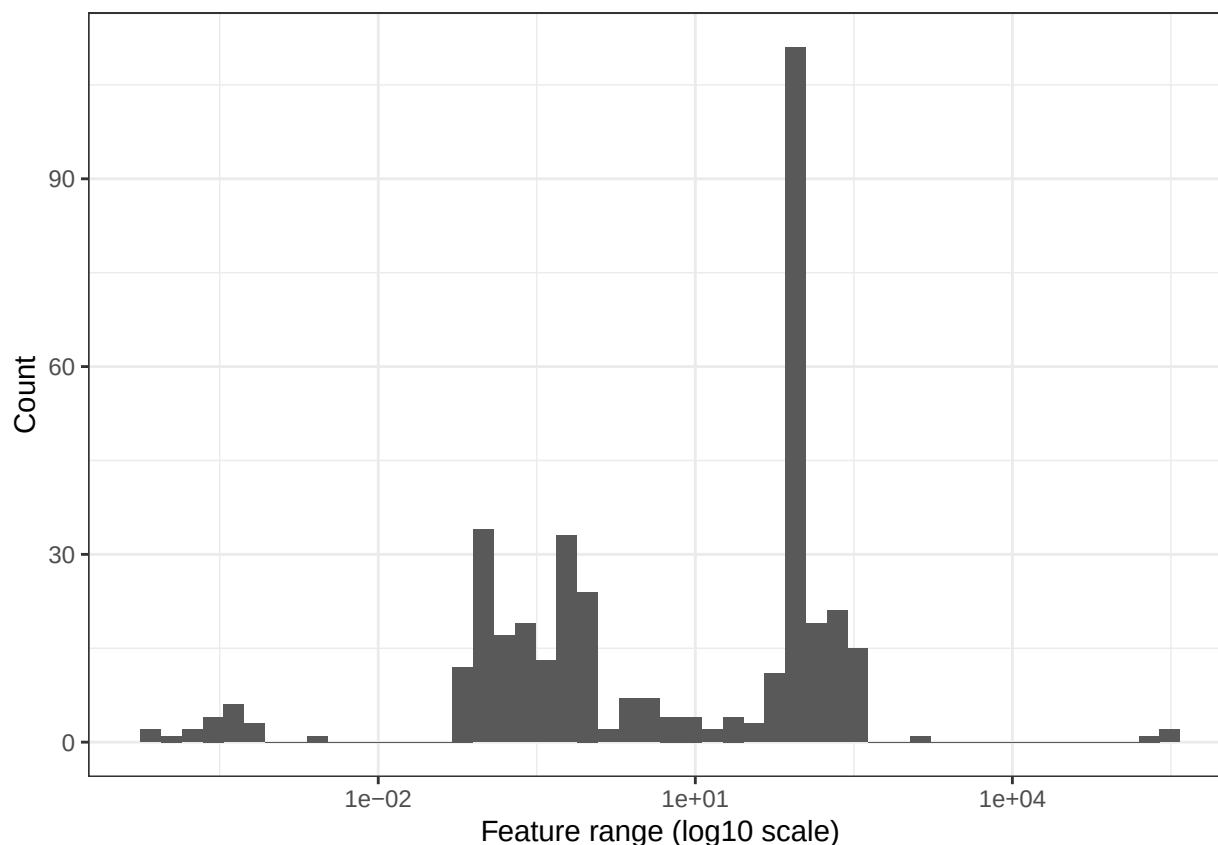
1.4 Feature distributions and scale

Analysis of the distributions and scales of the feature variables was conducted to assess their ranges, variability and differences in the scale.

Summary statistics of features including minimum, maximum, mean, and range were computed to identify the characteristics of each feature's scale.

```
## # A tibble: 385 x 5
##   Variable           Min           Max           Mean           Range
##   <chr>           <dbl>           <dbl>           <dbl>           <dbl>
## 1 feat_local_Entropy      0.226           3.67           3.12           3.45
## 2 feat_local_MolWeight 1181.           2314.          1936.          1133.
## 3 feat_local_AAtypes_Tiny      0              1              0.309          1
## 4 feat_local_AAtypes_Small    0.170           1              0.537          0.830
## 5 feat_local_AAtypes_Aliphatic 0              0.733          0.273          0.733
## 6 feat_local_AAtypes_Aromatic 0              0.486          0.115          0.486
## 7 feat_local_AAtypes_NonPolar 0.0580          1              0.542          0.942
## 8 feat_local_AAtypes_Polar      0              0.942          0.458          0.942
## 9 feat_local_AAtypes_Charged 0              0.719          0.224          0.719
## 10 feat_local_AAtypes_Basic    0              0.583          0.123          0.583
## # i 375 more rows
```

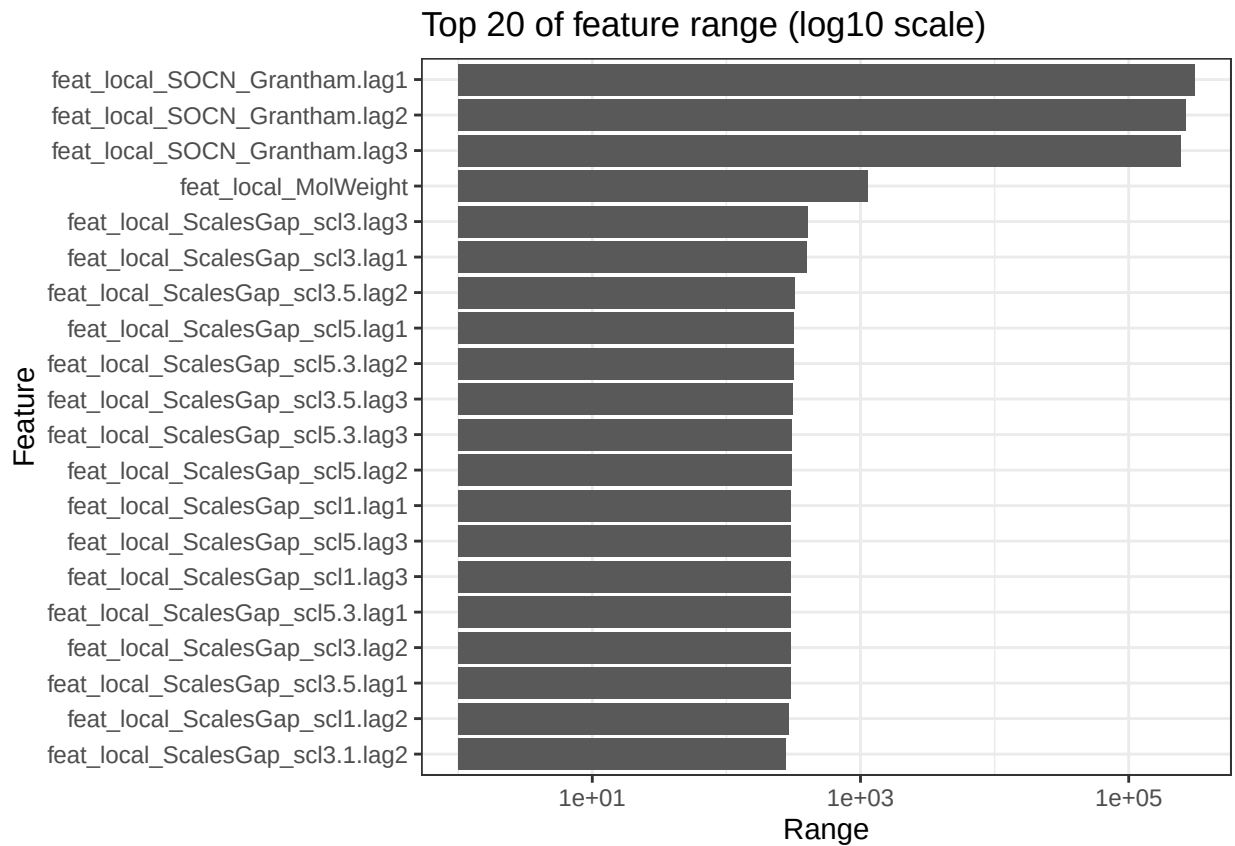
The histogram of feature ranges showed variations of their range which indicated heterogeneity in scale across features.



To take a closer look into the features scale, top 20 features with the largest range were extracted and plotted. The result showed highlighted features such as “feat_local_SOCN_Grantham.lag” and “feat_local_MolWeight” that could lead to information overload on the model due to wide spectrum of its range compared to other features.

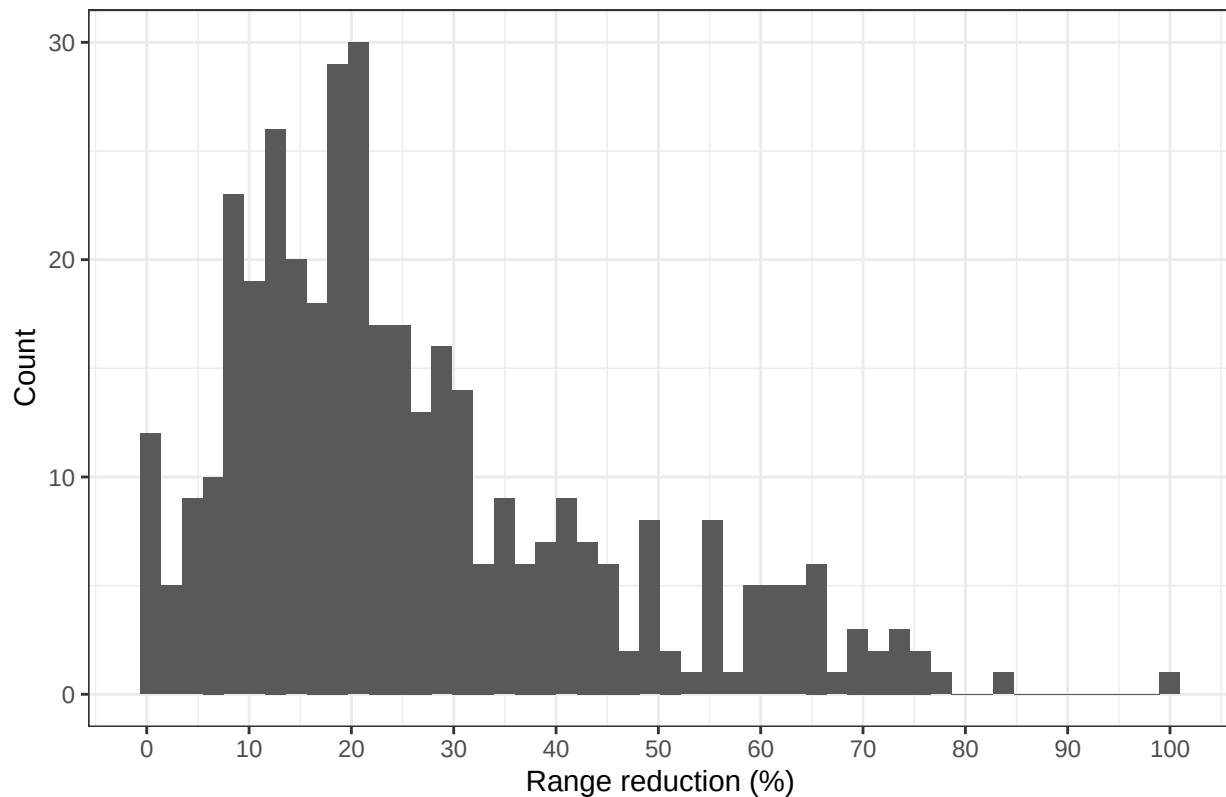
```
## # A tibble: 20 x 5
##   Variable           Min      Max      Mean  Range
##   <fct>           <dbl>    <dbl>    <dbl>  <dbl>
## 1 feat_local_SOCN_Grantham.lag1  8298.  318047.  141696.  309749.
## 2 feat_local_SOCN_Grantham.lag2  6870.  274117.  132470.  267246.
## 3 feat_local_SOCN_Grantham.lag3  6224.  248862.  120358.  242637.
## 4 feat_local_MolWeight          1181.   2314.   1936.   1133.
## 5 feat_local_ScalesGap_scl3.lag3   -173.   229.    4.41   402.
## 6 feat_local_ScalesGap_scl3.lag1   -169.   228.    4.01   397.
## 7 feat_local_ScalesGap_scl3.5.lag2 -250.    73.1    0.583   324.
## 8 feat_local_ScalesGap_scl5.lag1   -47.2   272.    0.850   319.
## 9 feat_local_ScalesGap_scl5.3.lag2 -247.    71.9    0.0227  319.
## 10 feat_local_ScalesGap_scl3.5.lag3 -251.    59.4   -0.651   310.
## 11 feat_local_ScalesGap_scl5.3.lag3 -246.    60.0    0.0811  306.
## 12 feat_local_ScalesGap_scl5.lag2   -35.5   270.    1.10   306.
## 13 feat_local_ScalesGap_scl1.lag1   -115.   189.    1.44   303.
## 14 feat_local_ScalesGap_scl5.lag3   -33.2   269.    0.840   303.
## 15 feat_local_ScalesGap_scl1.lag3   -133.   168.    2.47   302.
## 16 feat_local_ScalesGap_scl5.3.lag1 -248.    53.4   -0.112   301.
## 17 feat_local_ScalesGap_scl3.lag2   -71.8   229.    4.89   301.
## 18 feat_local_ScalesGap_scl3.5.lag1 -248.    50.7   -0.126   299.
```

## 19	feat_local_ScalesGap_scl1.lag2	-123.	169.	-2.06	292.
## 20	feat_local_ScalesGap_scl3.1.lag2	-186.	89.7	-0.985	276.



To check whether range difference is due to outliers, ranges were contrasted with truncated ranges. Removal of the upper/lower 0.1% of the data lead to the range reduction concentrated between 10 to 30%, indicating that most of the features show small change in scale when their tails are trimmed, while some show large reduction due to influence of their outliers.

Range reduction after truncating 0.1% upper/lower tails



The overview of feature distributions and scales show heterogeneous scales and high dimensionality. Hence, feature standardisation and dimensionality reduction will be performed to control the scale difference, redundancy and stability of the model.

1.5 EDA summary and implications for preprocessing

Based on the EDA, the following aspects of the dataset should be taken into account during preprocessing:

- Discard “Info_” variables that act primarily as identifiers / meta data.
- Split train and test data at the group level of “Info_group” to avoid data leakage.
- Apply simple median imputation for missing features.
- Standardise features to a common scale and conduct a dimensionality reduction such as PCA to reduce redundancy and increase stability to the model.

2. Data preprocessing and feature engineering

Note: It is expected that this section should result in no more than 400-ish words (possibly less; not counting the code blocks).

Add here your data pre-processing and feature selection/engineering steps. Your solution should include both your code (which markers must be able to run independently) and your rationale. In general, your steps should include:

- A short justification of which insight from your EDA is motivating a given data transformation.
- A code block implementing your pre-processing step.

In addition to the regular data transformations emerging from the insights you gathered during EDA, this section must contain the splitting off of a test set that you'll use later for final performance assessment (check the CW specs for details).

At the end of this section you should also include a short commentary on what changes the pre-processing has induced in your data (i.e., what did the data “look like” before this step vs. what it “looks like” afterwards).

You can use as many code blocks as needed, but don't overdo it (remember - everything needs to have a proper rationale). If you compare different pre-processing options, make sure to make it clear, and to indicate at the end which option you selected, and why.

Note: In this section you'll be basically defining which data transformations you will later incorporate into your predictive pipeline. When you build the pipeline in the next section, you'll encapsulate all those steps (together with the model) into a single object that you can fit and deploy as a single unit.

2.1 Split train and test data at the group level

Since a grouping structure exists in the dataset, the data is split into training and test sets at the group level before the sequential data preprocessing to minimise data leakage. However, the preprocessing in this section follow the predefined data structure and assumptions in the task specification. In particular, the target Class is converted into a factor and identifier columns that are not informative are excluded, leaving only numerical features to be used as inputs for modelling.

```
## Rows: 7,403
## Columns: 3
## $ Info_ID      <chr> "QLJ58341.1:11_491", "P06477.1:4_215.5", "ATQ39397.1:8_186"~
## $ Info_group   <fct> 61, 357, 467, 239, 9, 55, 150, 411, 34, 80, 48, 194, 143, 8~
## $ Class        <fct> epitope, non-epitope, non-epitope, non-epitope, non-epitope~
```

This split ensures that no Info_group appears in both training and test sets, preserving the independence of the final evaluation. From now onward, only the training data “df.train” will be used for data preprocessing and feature engineering.

2.2 Preprocessing recipe

Based on the EDA conducted, a recipe is created to build preprocessing workflow for data. The recipe selects the “feat_” columns as predictors, imputes missing values, standardise the predictors scales and conduct PCA for dimensionality reduction. For imputation, since only a small portion of missing values and outliers exist, median imputation was chosen as a more robust option than the mean, because it is less sensitive to extreme values. Moreover, although trimming or winsorisation to control outlier was considered, since their wide ranges seem to be natural due to their characteristics, it was not applied. Instead, standardisation and PCA was conducted.

```
##
## -- Recipe -----
##
## -- Inputs
## Number of variables by role
## outcome:      1
```

```
## predictor: 385
## group:      1
## id:         1
##
## -- Operations
## * Median imputation for: all_numeric_predictors()
## * Centering and scaling for: all_numeric_predictors()
## * PCA extraction with: all_numeric_predictors()
```

```
## # A tibble: 133 x 4
##   variable   type      role      source
##   <chr>     <list>   <chr>   <chr>
## 1 Info_ID   <chr [3]> id      original
## 2 Info_group <chr [3]> group  original
## 3 Class     <chr [3]> outcome original
## 4 PC001     <chr [2]> predictor derived
## 5 PC002     <chr [2]> predictor derived
## 6 PC003     <chr [2]> predictor derived
## 7 PC004     <chr [2]> predictor derived
## 8 PC005     <chr [2]> predictor derived
## 9 PC006     <chr [2]> predictor derived
## 10 PC007    <chr [2]> predictor derived
## # i 123 more rows
```

```
##                                     [,1]
## feat_local_Entropy_mean          -1.335045e-14
## feat_local_Entropy_sd             1.000000e+00
## feat_local_MolWeight_mean         -9.132850e-14
## feat_local_MolWeight_sd           1.000000e+00
## feat_local_AAtypes_Tiny_mean      -9.470530e-15
## feat_local_AAtypes_Tiny_sd        1.000000e+00
## feat_local_AAtypes_Small_mean     3.134176e-14
## feat_local_AAtypes_Small_sd       1.000000e+00
## feat_local_AAtypes_Aliphatic_mean -1.112616e-14
## feat_local_AAtypes_Aliphatic_sd   1.000000e+00
```

Further inspection on PCA's cumulative variance is conducted to identify the degree of contribution of each variable and ideal number of principal components (PCs) to obtain roughly 90 to 95% of the information.

```
## Importance of components:
##           PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  6.10978 5.59010 4.15483 3.82595 3.68797 3.65086 3.15539
## Proportion of Variance 0.09696 0.08117 0.04484 0.03802 0.03533 0.03462 0.02586
## Cumulative Proportion 0.09696 0.17813 0.22296 0.26098 0.29631 0.33093 0.35679
##           PC8      PC9      PC10     PC11     PC12     PC13     PC14
## Standard deviation  2.95945 2.81479 2.80714 2.54097 2.44754 2.32571 2.27734
## Proportion of Variance 0.02275 0.02058 0.02047 0.01677 0.01556 0.01405 0.01347
## Cumulative Proportion 0.37954 0.40012 0.42059 0.43736 0.45292 0.46697 0.48044
##           PC15     PC16     PC17     PC18     PC19     PC20     PC21
## Standard deviation  2.24239 2.18898 2.12080 2.09723 2.04453 2.0103 1.91980
## Proportion of Variance 0.01306 0.01245 0.01168 0.01142 0.01086 0.0105 0.00957
## Cumulative Proportion 0.49350 0.50595 0.51763 0.52905 0.53991 0.5504 0.55998
##           PC22     PC23     PC24     PC25     PC26     PC27     PC28
```

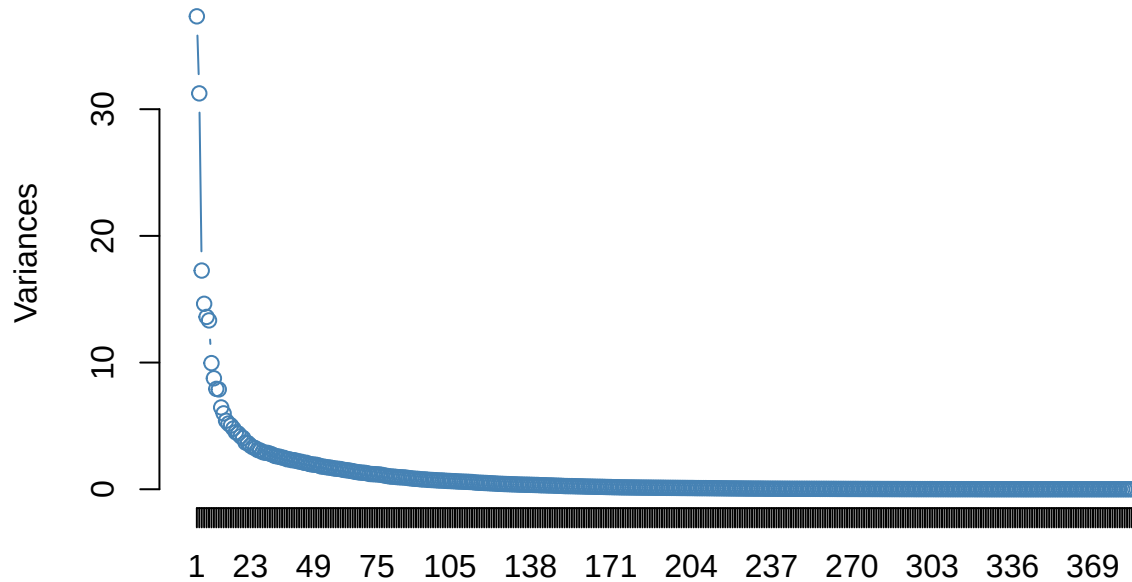
## Standard deviation	1.90629	1.85540	1.82041	1.80041	1.75860	1.74854	1.71239
## Proportion of Variance	0.00944	0.00894	0.00861	0.00842	0.00803	0.00794	0.00762
## Cumulative Proportion	0.56942	0.57836	0.58697	0.59539	0.60342	0.61136	0.61898
##	PC29	PC30	PC31	PC32	PC33	PC34	PC35
## Standard deviation	1.6989	1.69396	1.67517	1.65416	1.62185	1.61083	1.60094
## Proportion of Variance	0.0075	0.00745	0.00729	0.00711	0.00683	0.00674	0.00666
## Cumulative Proportion	0.6265	0.63393	0.64122	0.64832	0.65516	0.66190	0.66855
##	PC36	PC37	PC38	PC39	PC40	PC41	PC42
## Standard deviation	1.58101	1.5695	1.54058	1.53672	1.51566	1.51327	1.49589
## Proportion of Variance	0.00649	0.0064	0.00616	0.00613	0.00597	0.00595	0.00581
## Cumulative Proportion	0.67505	0.6814	0.68761	0.69374	0.69971	0.70566	0.71147
##	PC43	PC44	PC45	PC46	PC47	PC48	PC49
## Standard deviation	1.47926	1.46963	1.44645	1.44370	1.4154	1.40257	1.39372
## Proportion of Variance	0.00568	0.00561	0.00543	0.00541	0.0052	0.00511	0.00505
## Cumulative Proportion	0.71715	0.72276	0.72820	0.73361	0.7388	0.74392	0.74897
##	PC50	PC51	PC52	PC53	PC54	PC55	PC56
## Standard deviation	1.38872	1.37092	1.34174	1.33593	1.32271	1.31150	1.30431
## Proportion of Variance	0.00501	0.00488	0.00468	0.00464	0.00454	0.00447	0.00442
## Cumulative Proportion	0.75398	0.75886	0.76354	0.76817	0.77272	0.77718	0.78160
##	PC57	PC58	PC59	PC60	PC61	PC62	PC63
## Standard deviation	1.2868	1.27541	1.26934	1.26105	1.2403	1.22898	1.22067
## Proportion of Variance	0.0043	0.00423	0.00419	0.00413	0.0040	0.00392	0.00387
## Cumulative Proportion	0.7859	0.79013	0.79431	0.79844	0.8024	0.80636	0.81023
##	PC64	PC65	PC66	PC67	PC68	PC69	PC70
## Standard deviation	1.20817	1.1939	1.17433	1.16234	1.14926	1.14250	1.13623
## Proportion of Variance	0.00379	0.0037	0.00358	0.00351	0.00343	0.00339	0.00335
## Cumulative Proportion	0.81402	0.8177	0.82131	0.82482	0.82825	0.83164	0.83499
##	PC71	PC72	PC73	PC74	PC75	PC76	PC77
## Standard deviation	1.12166	1.09877	1.0933	1.09073	1.08377	1.07883	1.06282
## Proportion of Variance	0.00327	0.00314	0.0031	0.00309	0.00305	0.00302	0.00293
## Cumulative Proportion	0.83826	0.84140	0.8445	0.84759	0.85064	0.85366	0.85660
##	PC78	PC79	PC80	PC81	PC82	PC83	PC84
## Standard deviation	1.04323	1.02528	1.01045	0.99823	0.99563	0.98210	0.97480
## Proportion of Variance	0.00283	0.00273	0.00265	0.00259	0.00257	0.00251	0.00247
## Cumulative Proportion	0.85942	0.86216	0.86481	0.86740	0.86997	0.87248	0.87494
##	PC85	PC86	PC87	PC88	PC89	PC90	PC91
## Standard deviation	0.96972	0.96576	0.95442	0.93906	0.92735	0.91390	0.90662
## Proportion of Variance	0.00244	0.00242	0.00237	0.00229	0.00223	0.00217	0.00213
## Cumulative Proportion	0.87739	0.87981	0.88217	0.88446	0.88670	0.88887	0.89100
##	PC92	PC93	PC94	PC95	PC96	PC97	PC98
## Standard deviation	0.90110	0.88572	0.87923	0.8784	0.86097	0.85675	0.85066
## Proportion of Variance	0.00211	0.00204	0.00201	0.0020	0.00193	0.00191	0.00188
## Cumulative Proportion	0.89311	0.89515	0.89716	0.8992	0.90109	0.90299	0.90487
##	PC99	PC100	PC101	PC102	PC103	PC104	PC105
## Standard deviation	0.84504	0.83408	0.83052	0.82774	0.81677	0.8096	0.80624
## Proportion of Variance	0.00185	0.00181	0.00179	0.00178	0.00173	0.0017	0.00169
## Cumulative Proportion	0.90673	0.90854	0.91033	0.91211	0.91384	0.9155	0.91723
##	PC106	PC107	PC108	PC109	PC110	PC111	PC112
## Standard deviation	0.79974	0.79503	0.78351	0.77907	0.77218	0.76932	0.76450
## Proportion of Variance	0.00166	0.00164	0.00159	0.00158	0.00155	0.00154	0.00152
## Cumulative Proportion	0.91889	0.92053	0.92213	0.92370	0.92525	0.92679	0.92831
##	PC113	PC114	PC115	PC116	PC117	PC118	PC119
## Standard deviation	0.7590	0.75374	0.7348	0.72489	0.71510	0.71206	0.70423
## Proportion of Variance	0.0015	0.00148	0.0014	0.00136	0.00133	0.00132	0.00129

## Cumulative Proportion	0.9298	0.93128	0.9327	0.93405	0.93538	0.93669	0.93798
##	PC120	PC121	PC122	PC123	PC124	PC125	PC126
## Standard deviation	0.69933	0.69002	0.68353	0.67460	0.66623	0.65537	0.6509
## Proportion of Variance	0.00127	0.00124	0.00121	0.00118	0.00115	0.00112	0.0011
## Cumulative Proportion	0.93925	0.94049	0.94170	0.94288	0.94404	0.94515	0.9463
##	PC127	PC128	PC129	PC130	PC131	PC132	PC133
## Standard deviation	0.64478	0.64074	0.63498	0.62858	0.6215	0.6197	0.60827
## Proportion of Variance	0.00108	0.00107	0.00105	0.00103	0.0010	0.0010	0.00096
## Cumulative Proportion	0.94733	0.94840	0.94945	0.95047	0.9515	0.9525	0.95343
##	PC134	PC135	PC136	PC137	PC138	PC139	PC140
## Standard deviation	0.60481	0.60160	0.59613	0.59540	0.5885	0.58557	0.57951
## Proportion of Variance	0.00095	0.00094	0.00092	0.00092	0.0009	0.00089	0.00087
## Cumulative Proportion	0.95438	0.95532	0.95625	0.95717	0.9581	0.95896	0.95983
##	PC141	PC142	PC143	PC144	PC145	PC146	PC147
## Standard deviation	0.57809	0.56529	0.56127	0.55877	0.55076	0.54544	0.54414
## Proportion of Variance	0.00087	0.00083	0.00082	0.00081	0.00079	0.00077	0.00077
## Cumulative Proportion	0.96070	0.96153	0.96235	0.96316	0.96395	0.96472	0.96549
##	PC148	PC149	PC150	PC151	PC152	PC153	PC154
## Standard deviation	0.53124	0.52822	0.52136	0.51385	0.50256	0.49963	0.49642
## Proportion of Variance	0.00073	0.00072	0.00071	0.00069	0.00066	0.00065	0.00064
## Cumulative Proportion	0.96622	0.96694	0.96765	0.96834	0.96899	0.96964	0.97028
##	PC155	PC156	PC157	PC158	PC159	PC160	PC161
## Standard deviation	0.49455	0.48827	0.4821	0.47785	0.47319	0.46825	0.45737
## Proportion of Variance	0.00064	0.00062	0.0006	0.00059	0.00058	0.00057	0.00054
## Cumulative Proportion	0.97092	0.97154	0.9721	0.97273	0.97331	0.97388	0.97443
##	PC162	PC163	PC164	PC165	PC166	PC167	PC168
## Standard deviation	0.45551	0.45031	0.44619	0.4400	0.4373	0.43561	0.43000
## Proportion of Variance	0.00054	0.00053	0.00052	0.0005	0.0005	0.00049	0.00048
## Cumulative Proportion	0.97497	0.97549	0.97601	0.9765	0.9770	0.97750	0.97798
##	PC169	PC170	PC171	PC172	PC173	PC174	PC175
## Standard deviation	0.42330	0.41829	0.41428	0.40459	0.40140	0.39970	0.38743
## Proportion of Variance	0.00047	0.00045	0.00045	0.00043	0.00042	0.00041	0.00039
## Cumulative Proportion	0.97845	0.97890	0.97935	0.97977	0.98019	0.98061	0.98100
##	PC176	PC177	PC178	PC179	PC180	PC181	PC182
## Standard deviation	0.38473	0.38315	0.37980	0.37487	0.37367	0.36839	0.36557
## Proportion of Variance	0.00038	0.00038	0.00037	0.00037	0.00036	0.00035	0.00035
## Cumulative Proportion	0.98138	0.98176	0.98214	0.98250	0.98286	0.98322	0.98356
##	PC183	PC184	PC185	PC186	PC187	PC188	PC189
## Standard deviation	0.36294	0.35673	0.35640	0.35515	0.35037	0.34851	0.34529
## Proportion of Variance	0.00034	0.00033	0.00033	0.00033	0.00032	0.00032	0.00031
## Cumulative Proportion	0.98391	0.98424	0.98457	0.98489	0.98521	0.98553	0.98584
##	PC190	PC191	PC192	PC193	PC194	PC195	PC196
## Standard deviation	0.34335	0.3411	0.3384	0.33583	0.33138	0.33042	0.32770
## Proportion of Variance	0.00031	0.0003	0.0003	0.00029	0.00029	0.00028	0.00028
## Cumulative Proportion	0.98614	0.9865	0.9867	0.98704	0.98732	0.98761	0.98789
##	PC197	PC198	PC199	PC200	PC201	PC202	PC203
## Standard deviation	0.32714	0.32252	0.32026	0.31735	0.31515	0.31490	0.31084
## Proportion of Variance	0.00028	0.00027	0.00027	0.00026	0.00026	0.00026	0.00025
## Cumulative Proportion	0.98816	0.98843	0.98870	0.98896	0.98922	0.98948	0.98973
##	PC204	PC205	PC206	PC207	PC208	PC209	PC210
## Standard deviation	0.30866	0.30712	0.30129	0.29943	0.29776	0.29550	0.29465
## Proportion of Variance	0.00025	0.00024	0.00024	0.00023	0.00023	0.00023	0.00023
## Cumulative Proportion	0.98998	0.99022	0.99046	0.99069	0.99092	0.99115	0.99137
##	PC211	PC212	PC213	PC214	PC215	PC216	PC217

## Standard deviation	0.28843	0.28750	0.28652	0.2790	0.2770	0.27393	0.27193
## Proportion of Variance	0.00022	0.00021	0.00021	0.0002	0.0002	0.00019	0.00019
## Cumulative Proportion	0.99159	0.99180	0.99202	0.9922	0.9924	0.99261	0.99280
##	PC218	PC219	PC220	PC221	PC222	PC223	PC224
## Standard deviation	0.26536	0.26267	0.25891	0.25559	0.25239	0.24603	0.24502
## Proportion of Variance	0.00018	0.00018	0.00017	0.00017	0.00017	0.00016	0.00016
## Cumulative Proportion	0.99299	0.99317	0.99334	0.99351	0.99368	0.99383	0.99399
##	PC225	PC226	PC227	PC228	PC229	PC230	PC231
## Standard deviation	0.23957	0.23686	0.23443	0.23112	0.22695	0.22538	0.22344
## Proportion of Variance	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00013
## Cumulative Proportion	0.99414	0.99428	0.99443	0.99456	0.99470	0.99483	0.99496
##	PC232	PC233	PC234	PC235	PC236	PC237	PC238
## Standard deviation	0.22084	0.21965	0.21392	0.21107	0.20828	0.20600	0.20424
## Proportion of Variance	0.00013	0.00013	0.00012	0.00012	0.00011	0.00011	0.00011
## Cumulative Proportion	0.99509	0.99521	0.99533	0.99545	0.99556	0.99567	0.99578
##	PC239	PC240	PC241	PC242	PC243	PC244	PC245
## Standard deviation	0.20117	0.1998	0.1976	0.1943	0.1931	0.19014	0.18933
## Proportion of Variance	0.00011	0.0001	0.0001	0.0001	0.0001	0.00009	0.00009
## Cumulative Proportion	0.99588	0.9960	0.9961	0.9962	0.9963	0.99638	0.99647
##	PC246	PC247	PC248	PC249	PC250	PC251	PC252
## Standard deviation	0.18797	0.18732	0.18513	0.18405	0.18259	0.18189	0.17956
## Proportion of Variance	0.00009	0.00009	0.00009	0.00009	0.00009	0.00009	0.00008
## Cumulative Proportion	0.99656	0.99665	0.99674	0.99683	0.99692	0.99700	0.99709
##	PC253	PC254	PC255	PC256	PC257	PC258	PC259
## Standard deviation	0.17780	0.17653	0.17491	0.17346	0.17199	0.17126	0.16974
## Proportion of Variance	0.00008	0.00008	0.00008	0.00008	0.00008	0.00008	0.00007
## Cumulative Proportion	0.99717	0.99725	0.99733	0.99741	0.99748	0.99756	0.99763
##	PC260	PC261	PC262	PC263	PC264	PC265	PC266
## Standard deviation	0.16656	0.16615	0.16327	0.16138	0.15914	0.15649	0.15470
## Proportion of Variance	0.00007	0.00007	0.00007	0.00007	0.00007	0.00006	0.00006
## Cumulative Proportion	0.99771	0.99778	0.99785	0.99792	0.99798	0.99804	0.99811
##	PC267	PC268	PC269	PC270	PC271	PC272	PC273
## Standard deviation	0.15356	0.15240	0.14935	0.14907	0.14721	0.14679	0.14563
## Proportion of Variance	0.00006	0.00006	0.00006	0.00006	0.00006	0.00006	0.00006
## Cumulative Proportion	0.99817	0.99823	0.99829	0.99834	0.99840	0.99846	0.99851
##	PC274	PC275	PC276	PC277	PC278	PC279	PC280
## Standard deviation	0.14385	0.14208	0.14128	0.13897	0.13847	0.13661	0.13375
## Proportion of Variance	0.00005	0.00005	0.00005	0.00005	0.00005	0.00005	0.00005
## Cumulative Proportion	0.99857	0.99862	0.99867	0.99872	0.99877	0.99882	0.99886
##	PC281	PC282	PC283	PC284	PC285	PC286	PC287
## Standard deviation	0.13252	0.13002	0.12973	0.12759	0.12430	0.12275	0.12105
## Proportion of Variance	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004
## Cumulative Proportion	0.99891	0.99895	0.99900	0.99904	0.99908	0.99912	0.99916
##	PC288	PC289	PC290	PC291	PC292	PC293	PC294
## Standard deviation	0.11943	0.11573	0.11350	0.11069	0.10797	0.10756	0.10694
## Proportion of Variance	0.00004	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003
## Cumulative Proportion	0.99919	0.99923	0.99926	0.99929	0.99932	0.99935	0.99938
##	PC295	PC296	PC297	PC298	PC299	PC300	PC301
## Standard deviation	0.10443	0.10269	0.10097	0.10056	0.09950	0.09787	0.09704
## Proportion of Variance	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002
## Cumulative Proportion	0.99941	0.99944	0.99947	0.99949	0.99952	0.99954	0.99957
##	PC302	PC303	PC304	PC305	PC306	PC307	PC308
## Standard deviation	0.09530	0.09405	0.09204	0.09000	0.08910	0.08732	0.08688
## Proportion of Variance	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002

## Cumulative Proportion	0.99959	0.99961	0.99964	0.99966	0.99968	0.99970	0.99972
##	PC309	PC310	PC311	PC312	PC313	PC314	PC315
## Standard deviation	0.08291	0.08045	0.07958	0.07788	0.07763	0.07687	0.07609
## Proportion of Variance	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002
## Cumulative Proportion	0.99974	0.99975	0.99977	0.99978	0.99980	0.99982	0.99983
##	PC316	PC317	PC318	PC319	PC320	PC321	PC322
## Standard deviation	0.07436	0.07279	0.07146	0.06900	0.06877	0.06550	0.06311
## Proportion of Variance	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001
## Cumulative Proportion	0.99984	0.99986	0.99987	0.99988	0.99990	0.99991	0.99992
##	PC323	PC324	PC325	PC326	PC327	PC328	PC329
## Standard deviation	0.06014	0.05886	0.05194	0.04862	0.04717	0.04498	0.0437
## Proportion of Variance	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.0000
## Cumulative Proportion	0.99993	0.99994	0.99994	0.99995	0.99996	0.99996	1.0000
##	PC330	PC331	PC332	PC333	PC334	PC335	PC336
## Standard deviation	0.04246	0.0399	0.03569	0.03503	0.03301	0.02609	0.02576
## Proportion of Variance	0.00000	0.0000	0.00000	0.00000	0.00000	0.00000	0.00000
## Cumulative Proportion	0.99997	1.0000	0.99998	0.99998	0.99998	0.99999	0.99999
##	PC337	PC338	PC339	PC340	PC341	PC342	PC343
## Standard deviation	0.02249	0.02177	0.01954	0.01739	0.01719	0.01611	0.01565
## Proportion of Variance	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
## Cumulative Proportion	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999
##	PC344	PC345	PC346	PC347	PC348	PC349	PC350
## Standard deviation	0.01432	0.01384	0.01375	0.01369	0.01302	0.01297	0.01237
## Proportion of Variance	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
## Cumulative Proportion	0.99999	0.99999	0.99999	1.00000	1.00000	1.00000	1.00000
##	PC351	PC352	PC353	PC354	PC355	PC356	
## Standard deviation	0.01146	0.01072	0.0104	0.01028	0.009716	0.009101	
## Proportion of Variance	0.00000	0.00000	0.0000	0.00000	0.000000	0.000000	
## Cumulative Proportion	1.00000	1.00000	1.0000	1.00000	1.000000	1.000000	
##	PC357	PC358	PC359	PC360	PC361	PC362	
## Standard deviation	0.008852	0.008278	0.007711	0.007549	0.007042	0.006411	
## Proportion of Variance	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
## Cumulative Proportion	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	
##	PC363	PC364	PC365	PC366	PC367	PC368	
## Standard deviation	0.00615	0.005955	0.005788	0.005692	0.005006	0.004555	
## Proportion of Variance	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	
## Cumulative Proportion	1.00000	1.000000	1.000000	1.000000	1.000000	1.000000	
##	PC369	PC370	PC371	PC372	PC373	PC374	
## Standard deviation	0.003997	0.003606	0.003446	0.003112	0.00247	0.002274	
## Proportion of Variance	0.000000	0.000000	0.000000	0.000000	0.00000	0.000000	
## Cumulative Proportion	1.000000	1.000000	1.000000	1.000000	1.00000	1.000000	
##	PC375	PC376	PC377	PC378	PC379	PC380	
## Standard deviation	0.001755	0.001437	0.001264	0.001062	0.0009754	0.0008929	
## Proportion of Variance	0.000000	0.000000	0.000000	0.000000	0.0000000	0.0000000	
## Cumulative Proportion	1.000000	1.000000	1.000000	1.000000	1.0000000	1.0000000	
##	PC381	PC382	PC383	PC384	PC385		
## Standard deviation	0.000569	0.0002927	1.174e-14	2.713e-15	1.474e-15		
## Proportion of Variance	0.000000	0.0000000	0.000e+00	0.000e+00	0.000e+00		
## Cumulative Proportion	1.000000	1.0000000	1.000e+00	1.000e+00	1.000e+00		

Scree plot of PCA



```
##                               PC96      PC130
## Standard deviation      0.8609664 0.6285765
## Proportion of Variance 0.0019300 0.0010300
## Cumulative Proportion  0.9010900 0.9504700
```

Summary of the PCs and its scree plot indicate that roughly one third of the PCs account for most of the variability in the predictors. PC130's standard deviation tells us it's 0.63 dispersed in relation to the mean of the data, proportion of variance tells us the influence contributed and cumulative proportion tells us that 95% of the data is explained by PC130.

To wrap up, perform a quick data inspection to verify whether the initial recipe is properly implemented.

```
##      Info_ID      Info_group      Class      PC001
## Length:5925      29      : 506      epitope      :2328      Min.      : -22.4595
## Class :character  55      : 435      non-epitope:3597      1st Qu.: -3.9377
## Mode  :character  239      : 198                                     Median : -0.3129
##                                     43      : 196                                     Mean   :  0.0000
##                                     176      : 177                                     3rd Qu.:  3.6162
##                                     60      : 156                                     Max.   : 42.3874
##                                     (Other):4257
##      PC002      PC003      PC004      PC005
## Min.      : -26.4114      Min.      : -18.81285      Min.      : -30.7549      Min.      : -61.8233
## 1st Qu.: -2.7073      1st Qu.: -2.64626      1st Qu.: -2.2252      1st Qu.: -1.9921
## Median :  0.5099      Median :  0.07536      Median : -0.2064      Median :  0.3173
## Mean   :  0.0000      Mean   :  0.00000      Mean   :  0.0000      Mean   :  0.0000
## 3rd Qu.:  3.2408      3rd Qu.:  2.70911      3rd Qu.:  1.9323      3rd Qu.:  2.4002
```

```

## Max. : 18.8812 Max. : 19.76455 Max. : 44.7447 Max. : 10.6536
##
## PC006 PC007 PC008 PC009
## Min. :-18.158 Min. :-67.359843 Min. :-18.0652 Min. :-17.2899
## 1st Qu.: -2.027 1st Qu.: -1.568494 1st Qu.: -1.5382 1st Qu.: -1.6467
## Median : -0.179 Median : -0.003407 Median : -0.0665 Median : 0.2277
## Mean : 0.000 Mean : 0.000000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 1.821 3rd Qu.: 1.656934 3rd Qu.: 1.4713 3rd Qu.: 1.9169
## Max. : 76.137 Max. : 18.619010 Max. : 26.3744 Max. : 27.3209
##
## PC010 PC011 PC012 PC013
## Min. :-34.16382 Min. :-16.68970 Min. :-11.8435 Min. :-19.645
## 1st Qu.: -1.67244 1st Qu.: -1.39470 1st Qu.: -1.5833 1st Qu.: -1.463
## Median : -0.07992 Median : 0.07824 Median : -0.1127 Median : 0.113
## Mean : 0.00000 Mean : 0.00000 Mean : 0.0000 Mean : 0.000
## 3rd Qu.: 1.62228 3rd Qu.: 1.46734 3rd Qu.: 1.4503 3rd Qu.: 1.527
## Max. : 14.13823 Max. : 11.78551 Max. : 13.2114 Max. : 13.574
##
## PC014 PC015 PC016
## Min. :-10.03059 Min. :-11.07677 Min. :-12.089676
## 1st Qu.: -1.44128 1st Qu.: -1.46491 1st Qu.: -1.375437
## Median : -0.04344 Median : -0.03663 Median : -0.005393
## Mean : 0.00000 Mean : 0.00000 Mean : 0.000000
## 3rd Qu.: 1.28915 3rd Qu.: 1.40886 3rd Qu.: 1.293334
## Max. : 37.81612 Max. : 10.98515 Max. : 11.701356
##
## PC017 PC018 PC019
## Min. :-35.75472 Min. :-10.99307 Min. :-36.4522
## 1st Qu.: -1.24371 1st Qu.: -1.19413 1st Qu.: -1.2382
## Median : 0.03325 Median : 0.09597 Median : 0.0313
## Mean : 0.00000 Mean : 0.00000 Mean : 0.0000
## 3rd Qu.: 1.27494 3rd Qu.: 1.29210 3rd Qu.: 1.2135
## Max. : 10.35675 Max. : 44.70441 Max. : 9.7139
##
## PC020 PC021 PC022 PC023
## Min. :-10.314144 Min. :-15.83969 Min. :-8.4977 Min. :-8.11232
## 1st Qu.: -1.234516 1st Qu.: -1.17025 1st Qu.: -1.1261 1st Qu.: -1.12569
## Median : 0.007128 Median : 0.03734 Median : 0.1587 Median : -0.04856
## Mean : 0.000000 Mean : 0.00000 Mean : 0.0000 Mean : 0.00000
## 3rd Qu.: 1.230786 3rd Qu.: 1.18798 3rd Qu.: 1.2296 3rd Qu.: 1.04359
## Max. : 13.762348 Max. : 7.90860 Max. : 6.6985 Max. : 9.42901
##
## PC024 PC025 PC026 PC027
## Min. :-8.76711 Min. :-14.62606 Min. :-8.30835 Min. :-9.02698
## 1st Qu.: -1.16650 1st Qu.: -1.09845 1st Qu.: -1.10507 1st Qu.: -1.12135
## Median : -0.06932 Median : -0.02047 Median : 0.01995 Median : 0.03523
## Mean : 0.00000 Mean : 0.00000 Mean : 0.00000 Mean : 0.00000
## 3rd Qu.: 1.12067 3rd Qu.: 1.03622 3rd Qu.: 1.12645 3rd Qu.: 1.10312
## Max. : 10.40707 Max. : 12.16287 Max. : 8.22018 Max. : 7.36199
##
## PC028 PC029 PC030 PC031
## Min. :-7.787218 Min. :-10.33010 Min. :-9.538997 Min. :-8.41189
## 1st Qu.: -1.107304 1st Qu.: -1.05652 1st Qu.: -1.089717 1st Qu.: -0.99586
## Median : 0.004451 Median : 0.01509 Median : 0.008978 Median : -0.01236

```

##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000
##	3rd Qu.:	1.092811	3rd Qu.:	1.04377	3rd Qu.:	1.077703	3rd Qu.:	1.000006
##	Max.	: 7.636975	Max.	: 9.02077	Max.	: 7.321152	Max.	: 8.86140
##								
##	PC032		PC033		PC034		PC035	
##	Min.	:-11.48358	Min.	:-8.616504	Min.	:-9.00179	Min.	:-7.29384
##	1st Qu.:	-0.96309	1st Qu.:	-1.007986	1st Qu.:	-1.00755	1st Qu.:	-0.95837
##	Median	: 0.04456	Median	: 0.009969	Median	: 0.01151	Median	: 0.01422
##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000
##	3rd Qu.:	0.97655	3rd Qu.:	1.016065	3rd Qu.:	1.03094	3rd Qu.:	0.94524
##	Max.	: 16.59152	Max.	: 7.894811	Max.	: 7.45305	Max.	: 7.84484
##								
##	PC036		PC037		PC038		PC039	
##	Min.	:-8.554469	Min.	:-8.01665	Min.	:-6.62135	Min.	:-7.789447
##	1st Qu.:	-0.978226	1st Qu.:	-0.98201	1st Qu.:	-0.93199	1st Qu.:	-0.935836
##	Median	: 0.003976	Median	: 0.01239	Median	:-0.02226	Median	:-0.003077
##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000
##	3rd Qu.:	0.953497	3rd Qu.:	0.96884	3rd Qu.:	0.89463	3rd Qu.:	0.955115
##	Max.	: 7.747017	Max.	: 6.65249	Max.	:10.04067	Max.	: 6.737655
##								
##	PC040		PC041		PC042		PC043	
##	Min.	:-6.19537	Min.	:-7.53097	Min.	:-6.780806	Min.	:-6.029785
##	1st Qu.:	-0.93251	1st Qu.:	-0.92301	1st Qu.:	-0.935086	1st Qu.:	-0.879657
##	Median	: 0.01147	Median	:-0.01437	Median	:-0.001145	Median	: 0.007537
##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000
##	3rd Qu.:	0.94092	3rd Qu.:	0.89807	3rd Qu.:	0.928113	3rd Qu.:	0.917617
##	Max.	: 6.01586	Max.	: 7.94670	Max.	: 6.925766	Max.	: 8.004523
##								
##	PC044		PC045		PC046		PC047	
##	Min.	:-6.9897	Min.	:-5.88114	Min.	:-7.3177	Min.	:-6.44889
##	1st Qu.:	-0.8911	1st Qu.:	-0.92197	1st Qu.:	-0.8726	1st Qu.:	-0.89382
##	Median	:-0.0179	Median	: 0.00917	Median	: 0.0248	Median	: 0.01151
##	Mean	: 0.0000	Mean	: 0.000000	Mean	: 0.0000	Mean	: 0.000000
##	3rd Qu.:	0.8678	3rd Qu.:	0.88527	3rd Qu.:	0.8828	3rd Qu.:	0.87654
##	Max.	: 7.0217	Max.	: 6.38127	Max.	: 7.2149	Max.	: 7.56371
##								
##	PC048		PC049		PC050		PC051	
##	Min.	:-8.00412	Min.	:-8.380566	Min.	:-6.283528	Min.	:-6.16218
##	1st Qu.:	-0.84414	1st Qu.:	-0.844021	1st Qu.:	-0.860363	1st Qu.:	-0.77907
##	Median	:-0.03063	Median	:-0.007149	Median	: 0.005986	Median	: 0.01021
##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000
##	3rd Qu.:	0.82253	3rd Qu.:	0.895530	3rd Qu.:	0.867574	3rd Qu.:	0.77935
##	Max.	: 7.73511	Max.	: 6.865885	Max.	: 5.862544	Max.	: 7.87052
##								
##	PC052		PC053		PC054			
##	Min.	:-6.413444	Min.	:-7.673627	Min.	:-8.967973		
##	1st Qu.:	-0.815605	1st Qu.:	-0.771433	1st Qu.:	-0.765473		
##	Median	: 0.005149	Median	:-0.004659	Median	: 0.009416		
##	Mean	: 0.000000	Mean	: 0.000000	Mean	: 0.000000		
##	3rd Qu.:	0.816047	3rd Qu.:	0.772694	3rd Qu.:	0.752781		
##	Max.	: 7.958428	Max.	: 7.477226	Max.	: 6.645534		
##								
##	PC055		PC056		PC057		PC058	
##	Min.	:-5.745613	Min.	:-6.26688	Min.	:-5.563850	Min.	:-5.738250

##	1st Qu.: -0.825516	1st Qu.: -0.77486	1st Qu.: -0.803187	1st Qu.: -0.782878
##	Median : 0.001669	Median : 0.01369	Median : -0.008997	Median : -0.008086
##	Mean : 0.000000	Mean : 0.00000	Mean : 0.000000	Mean : 0.000000
##	3rd Qu.: 0.828843	3rd Qu.: 0.80340	3rd Qu.: 0.744384	3rd Qu.: 0.790627
##	Max. : 6.478510	Max. : 6.14020	Max. : 8.131174	Max. : 5.578124
##				
##	PC059	PC060	PC061	PC062
##	Min. : -5.56926	Min. : -5.27004	Min. : -5.759847	Min. : -5.9058
##	1st Qu.: -0.75701	1st Qu.: -0.77967	1st Qu.: -0.765320	1st Qu.: -0.7792
##	Median : -0.02323	Median : -0.02053	Median : -0.006796	Median : -0.0174
##	Mean : 0.00000	Mean : 0.00000	Mean : 0.000000	Mean : 0.0000
##	3rd Qu.: 0.76655	3rd Qu.: 0.78050	3rd Qu.: 0.732724	3rd Qu.: 0.7562
##	Max. : 5.86663	Max. : 6.56049	Max. : 5.835980	Max. : 7.8060
##				
##	PC063	PC064	PC065	PC066
##	Min. : -5.46716	Min. : -5.05202	Min. : -6.27180	Min. : -4.99921
##	1st Qu.: -0.75308	1st Qu.: -0.74881	1st Qu.: -0.71240	1st Qu.: -0.71454
##	Median : -0.01871	Median : -0.00296	Median : 0.01882	Median : 0.03159
##	Mean : 0.00000	Mean : 0.00000	Mean : 0.00000	Mean : 0.00000
##	3rd Qu.: 0.72177	3rd Qu.: 0.75212	3rd Qu.: 0.70323	3rd Qu.: 0.71917
##	Max. : 5.38707	Max. : 5.16340	Max. : 6.27676	Max. : 6.49675
##				
##	PC067	PC068	PC069	PC070
##	Min. : -4.40763	Min. : -4.9893	Min. : -4.78499	Min. : -6.653073
##	1st Qu.: -0.71941	1st Qu.: -0.7328	1st Qu.: -0.69871	1st Qu.: -0.671083
##	Median : -0.04304	Median : -0.0404	Median : -0.01419	Median : 0.001019
##	Mean : 0.00000	Mean : 0.0000	Mean : 0.00000	Mean : 0.000000
##	3rd Qu.: 0.69284	3rd Qu.: 0.6954	3rd Qu.: 0.68315	3rd Qu.: 0.691532
##	Max. : 6.11475	Max. : 5.1391	Max. : 6.62951	Max. : 11.933726
##				
##	PC071	PC072	PC073	
##	Min. : -6.324427	Min. : -4.4705387	Min. : -5.702973	
##	1st Qu.: -0.678936	1st Qu.: -0.6775776	1st Qu.: -0.649983	
##	Median : -0.009161	Median : -0.0007739	Median : 0.001929	
##	Mean : 0.000000	Mean : 0.0000000	Mean : 0.000000	
##	3rd Qu.: 0.699730	3rd Qu.: 0.6621868	3rd Qu.: 0.672023	
##	Max. : 5.141476	Max. : 5.1746848	Max. : 5.049805	
##				
##	PC074	PC075	PC076	
##	Min. : -5.555696	Min. : -5.0444106	Min. : -5.947738	
##	1st Qu.: -0.645924	1st Qu.: -0.6805553	1st Qu.: -0.664748	
##	Median : 0.008605	Median : 0.0005411	Median : -0.005352	
##	Mean : 0.000000	Mean : 0.0000000	Mean : 0.000000	
##	3rd Qu.: 0.675525	3rd Qu.: 0.6617257	3rd Qu.: 0.677282	
##	Max. : 4.533436	Max. : 6.3359075	Max. : 4.735672	
##				
##	PC077	PC078	PC079	PC080
##	Min. : -5.100300	Min. : -5.01144	Min. : -5.87575	Min. : -5.36017
##	1st Qu.: -0.640076	1st Qu.: -0.62924	1st Qu.: -0.64543	1st Qu.: -0.61420
##	Median : -0.003777	Median : -0.02668	Median : -0.01267	Median : 0.02185
##	Mean : 0.000000	Mean : 0.00000	Mean : 0.00000	Mean : 0.00000
##	3rd Qu.: 0.647319	3rd Qu.: 0.64367	3rd Qu.: 0.64047	3rd Qu.: 0.64497
##	Max. : 5.438038	Max. : 4.23934	Max. : 4.49027	Max. : 3.39670
##				

##	PC081	PC082	PC083	PC084
##	Min. : -3.67585	Min. : -4.363905	Min. : -4.0048	Min. : -4.520192
##	1st Qu.: -0.63369	1st Qu.: -0.605635	1st Qu.: -0.6033	1st Qu.: -0.609246
##	Median : -0.03325	Median : -0.002179	Median : 0.0137	Median : -0.008057
##	Mean : 0.00000	Mean : 0.000000	Mean : 0.0000	Mean : 0.000000
##	3rd Qu.: 0.61820	3rd Qu.: 0.633428	3rd Qu.: 0.6057	3rd Qu.: 0.582378
##	Max. : 5.33666	Max. : 4.584212	Max. : 4.8298	Max. : 4.128592
##				
##	PC085	PC086	PC087	PC088
##	Min. : -5.066785	Min. : -5.016524	Min. : -4.698571	Min. : -4.09583
##	1st Qu.: -0.576587	1st Qu.: -0.600709	1st Qu.: -0.578358	1st Qu.: -0.57502
##	Median : -0.004881	Median : -0.001992	Median : -0.003488	Median : 0.01945
##	Mean : 0.000000	Mean : 0.000000	Mean : 0.000000	Mean : 0.000000
##	3rd Qu.: 0.589463	3rd Qu.: 0.591743	3rd Qu.: 0.572846	3rd Qu.: 0.58511
##	Max. : 6.545981	Max. : 5.839788	Max. : 6.655795	Max. : 4.15060
##				
##	PC089	PC090	PC091	PC092
##	Min. : -3.47198	Min. : -5.43534	Min. : -5.099246	Min. : -3.88960
##	1st Qu.: -0.58098	1st Qu.: -0.54548	1st Qu.: -0.548069	1st Qu.: -0.53104
##	Median : 0.01167	Median : -0.01712	Median : -0.004191	Median : 0.00342
##	Mean : 0.00000	Mean : 0.00000	Mean : 0.000000	Mean : 0.00000
##	3rd Qu.: 0.55960	3rd Qu.: 0.55073	3rd Qu.: 0.562231	3rd Qu.: 0.54228
##	Max. : 4.43991	Max. : 6.12574	Max. : 10.966838	Max. : 3.63852
##				
##	PC093	PC094	PC095	PC096
##	Min. : -3.92348	Min. : -3.92105	Min. : -4.757172	Min. : -8.9381238
##	1st Qu.: -0.54197	1st Qu.: -0.51935	1st Qu.: -0.514426	1st Qu.: -0.4981790
##	Median : 0.01804	Median : 0.02025	Median : 0.008521	Median : 0.0001525
##	Mean : 0.00000	Mean : 0.00000	Mean : 0.000000	Mean : 0.0000000
##	3rd Qu.: 0.55954	3rd Qu.: 0.53497	3rd Qu.: 0.545576	3rd Qu.: 0.5033263
##	Max. : 3.50441	Max. : 4.11880	Max. : 3.786733	Max. : 4.7090037
##				
##	PC097	PC098	PC099	PC100
##	Min. : -5.367604	Min. : -4.3771	Min. : -4.076728	Min. : -4.5243771
##	1st Qu.: -0.509543	1st Qu.: -0.5082	1st Qu.: -0.514014	1st Qu.: -0.5185659
##	Median : 0.008917	Median : -0.0173	Median : -0.007267	Median : -0.0004089
##	Mean : 0.000000	Mean : 0.0000	Mean : 0.000000	Mean : 0.0000000
##	3rd Qu.: 0.520196	3rd Qu.: 0.5101	3rd Qu.: 0.513058	3rd Qu.: 0.5194029
##	Max. : 3.610958	Max. : 5.1465	Max. : 5.316226	Max. : 4.9534241
##				
##	PC101	PC102	PC103	PC104
##	Min. : -4.237707	Min. : -3.99934	Min. : -4.410634	Min. : -3.873699
##	1st Qu.: -0.530954	1st Qu.: -0.51994	1st Qu.: -0.492634	1st Qu.: -0.495612
##	Median : -0.002839	Median : -0.01182	Median : 0.009334	Median : -0.007537
##	Mean : 0.000000	Mean : 0.00000	Mean : 0.000000	Mean : 0.0000000
##	3rd Qu.: 0.513206	3rd Qu.: 0.51414	3rd Qu.: 0.511376	3rd Qu.: 0.505634
##	Max. : 4.148925	Max. : 4.17933	Max. : 4.064287	Max. : 4.314925
##				
##	PC105	PC106	PC107	PC108
##	Min. : -4.536608	Min. : -4.04960	Min. : -8.818016	Min. : -3.824933
##	1st Qu.: -0.491847	1st Qu.: -0.48502	1st Qu.: -0.464414	1st Qu.: -0.488428
##	Median : -0.003855	Median : 0.01999	Median : -0.009634	Median : -0.004048
##	Mean : 0.000000	Mean : 0.00000	Mean : 0.000000	Mean : 0.0000000
##	3rd Qu.: 0.487426	3rd Qu.: 0.50164	3rd Qu.: 0.467757	3rd Qu.: 0.489558

```

## Max. : 4.832510 Max. : 3.99065 Max. : 7.621567 Max. : 3.227025
##
## PC109 PC110 PC111
## Min. :-4.514296 Min. :-2.979393 Min. :-4.190194
## 1st Qu.:-0.470983 1st Qu.:-0.495908 1st Qu.:-0.469903
## Median :-0.009333 Median : 0.001122 Median : 0.007822
## Mean : 0.000000 Mean : 0.000000 Mean : 0.000000
## 3rd Qu.: 0.481811 3rd Qu.: 0.481651 3rd Qu.: 0.480752
## Max. : 3.862737 Max. : 6.393579 Max. : 3.421513
##
## PC112 PC113 PC114
## Min. :-3.464607 Min. :-3.218813 Min. :-5.73401
## 1st Qu.:-0.465387 1st Qu.:-0.482443 1st Qu.:-0.46351
## Median :-0.000209 Median : 0.001043 Median :-0.00969
## Mean : 0.000000 Mean : 0.000000 Mean : 0.000000
## 3rd Qu.: 0.469120 3rd Qu.: 0.464575 3rd Qu.: 0.47115
## Max. : 6.972206 Max. : 4.419444 Max. : 3.04368
##
## PC115 PC116 PC117
## Min. :-17.694387 Min. :-3.404453 Min. :-10.26419
## 1st Qu.: -0.410592 1st Qu.:-0.440913 1st Qu.: -0.41891
## Median : 0.005211 Median : 0.002647 Median : -0.01988
## Mean : 0.000000 Mean : 0.000000 Mean : 0.000000
## 3rd Qu.: 0.430179 3rd Qu.: 0.434627 3rd Qu.: 0.40370
## Max. : 3.081211 Max. :11.681313 Max. : 11.32728
##
## PC118 PC119 PC120
## Min. :-16.93149 Min. :-11.590038 Min. :-3.772782
## 1st Qu.: -0.40666 1st Qu.: -0.422232 1st Qu.:-0.367781
## Median : 0.01536 Median : 0.002985 Median : 0.003587
## Mean : 0.000000 Mean : 0.000000 Mean : 0.000000
## 3rd Qu.: 0.41663 3rd Qu.: 0.432069 3rd Qu.: 0.359847
## Max. : 3.37677 Max. : 2.641935 Max. :18.132226
##
## PC121 PC122 PC123 PC124
## Min. :-3.25979 Min. :-6.660468 Min. :-2.70236 Min. :-6.746984
## 1st Qu.:-0.42243 1st Qu.:-0.417601 1st Qu.:-0.41426 1st Qu.:-0.399680
## Median :-0.01205 Median :-0.001827 Median : 0.00824 Median :-0.008915
## Mean : 0.000000 Mean : 0.000000 Mean : 0.000000 Mean : 0.000000
## 3rd Qu.: 0.39492 3rd Qu.: 0.410984 3rd Qu.: 0.40353 3rd Qu.: 0.401181
## Max. : 5.34612 Max. : 6.066260 Max. : 3.08693 Max. : 8.061654
##
## PC125 PC126 PC127
## Min. :-6.85319 Min. :-1.020e+01 Min. :-6.591285
## 1st Qu.:-0.39449 1st Qu.:-3.905e-01 1st Qu.:-0.387860
## Median :-0.00135 Median : 2.275e-04 Median :-0.003439
## Mean : 0.000000 Mean : 0.000e+00 Mean : 0.000000
## 3rd Qu.: 0.40239 3rd Qu.: 3.840e-01 3rd Qu.: 0.392378
## Max. : 4.01874 Max. : 3.328e+00 Max. : 2.713185
##
## PC128 PC129 PC130
## Min. :-10.425229 Min. :-3.869281 Min. :-2.576631
## 1st Qu.: -0.380726 1st Qu.:-0.403785 1st Qu.:-0.402934
## Median : -0.003867 Median :-0.001043 Median : 0.007119

```

```
## Mean      : 0.000000    Mean      : 0.000000    Mean      : 0.000000
## 3rd Qu.: 0.391001    3rd Qu.: 0.400783    3rd Qu.: 0.392879
## Max.      : 2.934801    Max.      : 3.052637    Max.      : 5.415646
##
```

```
## tibble [5,925 x 133] (S3: tbl_df/tbl/data.frame)
```

```
## $ Info_ID      : chr [1:5925] "QLJ58341.1:11_491" "P06477.1:4_215.5" "ABD75543.1:4_2589" "P12823.1:8_8"
## $ Info_group: Factor w/ 547 levels "2","3","4","5",...: 54 321 215 48 139 371 29 41 178 133 ...
## $ Class        : Factor w/ 2 levels "epitope","non-epitope": 1 2 2 1 2 2 2 1 2 ...
## $ PC001        : num [1:5925] -4.548 3.921 -1.228 -5.704 0.317 ...
## $ PC002        : num [1:5925] 1.001 -2.004 5.197 -1.899 -0.199 ...
## $ PC003        : num [1:5925] -0.915 2.205 3.491 4.508 -3.375 ...
## $ PC004        : num [1:5925] 0.766 -2.215 -2.913 -3.974 5.889 ...
## $ PC005        : num [1:5925] 2.244 -2.375 1.816 -0.583 0.389 ...
## $ PC006        : num [1:5925] 1.012 1.099 0.275 -0.987 0.099 ...
## $ PC007        : num [1:5925] -0.5664 0.0138 -2.6427 -0.1588 3.2016 ...
## $ PC008        : num [1:5925] -0.186 0.7589 0.7871 0.0187 -2.6198 ...
## $ PC009        : num [1:5925] 1.49 3.02 -1.21 1.1 2.74 ...
## $ PC010        : num [1:5925] 1.614 0.268 -1.926 0.388 -3.112 ...
## $ PC011        : num [1:5925] -1.3695 -1.0116 0.6798 0.0601 3.2587 ...
## $ PC012        : num [1:5925] -0.139 0.145 2.005 -4.518 -1.4 ...
## $ PC013        : num [1:5925] 0.558 0.871 2.35 -2.558 2.125 ...
## $ PC014        : num [1:5925] -2.611 -1.606 1.782 1.907 -0.418 ...
## $ PC015        : num [1:5925] 1.621 -0.701 1.161 -1.037 1.53 ...
## $ PC016        : num [1:5925] -3.244 -0.317 -1.584 -1.474 0.452 ...
## $ PC017        : num [1:5925] 1.89 -1.86 1.38 -1.51 -1.05 ...
## $ PC018        : num [1:5925] 2.105 -0.379 -2.543 -0.315 2.505 ...
## $ PC019        : num [1:5925] -1.754 -0.493 -0.404 -0.747 -2.145 ...
## $ PC020        : num [1:5925] -0.652 0.38 -3.107 -1.198 0.584 ...
## $ PC021        : num [1:5925] 2.336 -1.3006 0.0202 1.4497 0.6716 ...
## $ PC022        : num [1:5925] -4.283 -0.913 -0.926 -2.249 0.202 ...
## $ PC023        : num [1:5925] -1.884 -0.247 -0.76 -1.941 -1.459 ...
## $ PC024        : num [1:5925] 0.702 -0.665 -1.09 0.561 1.193 ...
## $ PC025        : num [1:5925] 0.682 0.576 2.782 2.476 -0.28 ...
## $ PC026        : num [1:5925] 2.897 0.623 -0.796 -0.503 1.955 ...
## $ PC027        : num [1:5925] -1.334 0.46 -0.464 -0.338 0.533 ...
## $ PC028        : num [1:5925] 0.0232 -0.4013 3.5418 -1.4644 -0.3519 ...
## $ PC029        : num [1:5925] -0.986 1.472 0.264 0.703 2.777 ...
## $ PC030        : num [1:5925] 1.6204 0.0482 -1.5455 0.6332 -0.0905 ...
## $ PC031        : num [1:5925] 0.392 -0.165 1.221 1.289 0.31 ...
## $ PC032        : num [1:5925] 1.747 -0.692 1.007 0.831 0.269 ...
## $ PC033        : num [1:5925] -1.3182 -0.4042 0.0675 -1.4207 -0.5224 ...
## $ PC034        : num [1:5925] -2.0106 -0.5077 -0.0928 -2.3026 -2.0606 ...
## $ PC035        : num [1:5925] 0.79706 0.73964 0.00483 -0.23097 -1.32633 ...
## $ PC036        : num [1:5925] -0.282 0.434 2.199 0.304 1.001 ...
## $ PC037        : num [1:5925] -3.958 0.215 0.619 -0.736 0.857 ...
## $ PC038        : num [1:5925] 0.107 -0.295 -0.31 1.234 -0.851 ...
## $ PC039        : num [1:5925] 2.389 -0.408 3.046 -0.808 0.351 ...
## $ PC040        : num [1:5925] 0.0137 -0.6528 0.2824 -1.0792 -2.4924 ...
## $ PC041        : num [1:5925] -0.508 0.639 -1.922 -1.334 -0.667 ...
## $ PC042        : num [1:5925] 0.515 -0.128 -1.165 -0.321 -0.187 ...
## $ PC043        : num [1:5925] 0.182 -0.219 0.254 -0.25 -1.948 ...
## $ PC044        : num [1:5925] 0.4774 0.0324 -2.0526 -0.4994 0.3166 ...
## $ PC045        : num [1:5925] -0.171 0.333 1.031 0.292 -0.777 ...
```

```

## $ PC046 : num [1:5925] 0.287937 -0.611904 -0.903227 0.982493 -0.000687 ...
## $ PC047 : num [1:5925] -0.384 0.5067 1.6773 -0.0349 0.2706 ...
## $ PC048 : num [1:5925] -1.071 -0.192 -0.155 0.943 -0.501 ...
## $ PC049 : num [1:5925] -0.8606 -0.1857 0.0564 0.0576 0.5027 ...
## $ PC050 : num [1:5925] -0.419 0.0204 -0.1572 0.8689 0.5278 ...
## $ PC051 : num [1:5925] 1.606 -0.224 0.207 1.012 0.579 ...
## $ PC052 : num [1:5925] 1.048 0.108 0.444 -0.993 -0.249 ...
## $ PC053 : num [1:5925] -2.804 0.426 1.349 -0.354 -1.207 ...
## $ PC054 : num [1:5925] 0.797 0.216 0.191 0.793 0.454 ...
## $ PC055 : num [1:5925] -0.422 0.66 -0.865 0.708 1.028 ...
## $ PC056 : num [1:5925] 0.265 -0.396 0.258 -0.371 0.109 ...
## $ PC057 : num [1:5925] -0.6601 -0.0876 -0.3278 0.8631 -0.5467 ...
## $ PC058 : num [1:5925] 1.0855 -0.0595 -0.314 -0.8945 0.3077 ...
## $ PC059 : num [1:5925] 0.444 0.217 0.617 -0.354 1.228 ...
## $ PC060 : num [1:5925] -0.804 0.921 -0.705 0.222 0.415 ...
## $ PC061 : num [1:5925] -1.18 -0.181 0.619 1.499 1.301 ...
## $ PC062 : num [1:5925] 1.41512 -0.76254 1.10281 0.00654 -0.47108 ...
## $ PC063 : num [1:5925] -1.335 -0.696 1.479 1.056 0.635 ...
## $ PC064 : num [1:5925] -1.7006 0.3403 0.5119 -0.0885 -0.7893 ...
## $ PC065 : num [1:5925] 1.013 0.536 -1.133 -1.94 0.35 ...
## $ PC066 : num [1:5925] -1.4839 -0.4006 -0.7576 0.4045 -0.0519 ...
## $ PC067 : num [1:5925] -1.211 0.273 -1.597 1.252 0.456 ...
## $ PC068 : num [1:5925] -0.9 0.75 -0.5 -0.991 0.392 ...
## $ PC069 : num [1:5925] 0.4613 -0.0639 -0.0555 -1.653 -0.189 ...
## $ PC070 : num [1:5925] 1.281 -0.437 0.252 0.692 -0.273 ...
## $ PC071 : num [1:5925] 1.049 0.232 -0.622 0.881 -0.615 ...
## $ PC072 : num [1:5925] -1.406 -0.205 -0.576 0.675 0.397 ...
## $ PC073 : num [1:5925] -1.326 0.294 1.371 1.43 0.461 ...
## $ PC074 : num [1:5925] 0.2774 -0.0218 1.1269 0.0529 -0.1307 ...
## $ PC075 : num [1:5925] -0.78 0.692 -2.187 -1.115 0.218 ...
## $ PC076 : num [1:5925] -0.903 -0.234 -0.525 -0.147 1.763 ...
## $ PC077 : num [1:5925] 0.6301 0.3104 -0.566 1.2834 0.0534 ...
## $ PC078 : num [1:5925] 0.0251 0.7214 0.102 -0.3175 0.208 ...
## $ PC079 : num [1:5925] 0.395 1.314 -2.247 -0.79 -0.476 ...
## $ PC080 : num [1:5925] 2.708 0.18 0.296 0.598 -1.38 ...
## $ PC081 : num [1:5925] -1.459 -0.573 -0.874 -0.288 -0.938 ...
## $ PC082 : num [1:5925] -0.0357 1.0495 2.8616 1.2915 -0.4417 ...
## $ PC083 : num [1:5925] 1.142 -0.223 0.297 -0.584 0.927 ...
## $ PC084 : num [1:5925] 1.1898 0.3979 0.0674 -0.8342 0.8198 ...
## $ PC085 : num [1:5925] 0.0864 0.0788 0.143 1.031 -0.5933 ...
## $ PC086 : num [1:5925] 0.824 0.386 -2.201 -1.398 1.679 ...
## $ PC087 : num [1:5925] 0.2274 -0.3869 0.0683 -1.356 -0.0829 ...
## $ PC088 : num [1:5925] 0.744 0.341 0.175 -0.109 -1.521 ...
## $ PC089 : num [1:5925] 0.4513 -0.0591 -0.8845 -0.6287 -1.0274 ...
## $ PC090 : num [1:5925] -0.5495 -0.0425 -0.5024 0.231 -0.104 ...
## $ PC091 : num [1:5925] -0.979 -0.185 -0.749 -0.451 -0.657 ...
## $ PC092 : num [1:5925] 0.386 0.532 3.392 -0.287 -0.246 ...
## $ PC093 : num [1:5925] -0.193 0.197 -1.815 0.109 -0.123 ...
## $ PC094 : num [1:5925] -0.0465 0.227 -0.8429 -0.1489 -0.744 ...
## $ PC095 : num [1:5925] -0.427 -0.112 0.945 1.188 -0.549 ...
## $ PC096 : num [1:5925] 0.4997 -0.0799 2.5622 0.0488 -0.1138 ...
## [list output truncated]
## # A tibble: 6 x 133
## Info_ID Info_group Class PC001 PC002 PC003 PC004 PC005 PC006 PC007

```

```
##   <chr>      <fct>      <fct> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 QLJ58341.~ 61      epit~ -4.55  1.00 -0.915 0.766 2.24  1.01 -0.566
## 2 P06477.1:~ 357     non~~  3.92 -2.00  2.21 -2.22 -2.38  1.10  0.0138
## 3 ABD75543.~ 239     non~~ -1.23  5.20  3.49 -2.91  1.82  0.275 -2.64
## 4 P12823.1:~ 55      epit~ -5.70 -1.90  4.51 -3.97 -0.583 -0.987 -0.159
## 5 AGV08492.~ 150     non~~  0.317 -0.199 -3.37  5.89  0.389  0.0990 3.20
## 6 AAK32189.~ 411     non~~ 10.2   1.59 -3.64 -2.25  4.00 -0.312 3.09
## # i 123 more variables: PC008 <dbl>, PC009 <dbl>, PC010 <dbl>, PC011 <dbl>,
## #   PC012 <dbl>, PC013 <dbl>, PC014 <dbl>, PC015 <dbl>, PC016 <dbl>,
## #   PC017 <dbl>, PC018 <dbl>, PC019 <dbl>, PC020 <dbl>, PC021 <dbl>,
## #   PC022 <dbl>, PC023 <dbl>, PC024 <dbl>, PC025 <dbl>, PC026 <dbl>,
## #   PC027 <dbl>, PC028 <dbl>, PC029 <dbl>, PC030 <dbl>, PC031 <dbl>,
## #   PC032 <dbl>, PC033 <dbl>, PC034 <dbl>, PC035 <dbl>, PC036 <dbl>,
## #   PC037 <dbl>, PC038 <dbl>, PC039 <dbl>, PC040 <dbl>, PC041 <dbl>, ...

## [1] 0
```

Overall, the recipe function transforms the original high-dimensional dataset into a standardised version ideal for modelling. Now, with the analysis done in data preprocessing, pipelines using PCA thresholds of approximately 90% and 95% cumulative variance will be compared with grouped cross-validation and F1-score then the configuration with better performance will be selected as the final model.

3. Modelling

Note: It is expected that this section should result in no more than 400-ish words (possibly less; not counting the code blocks).

Add here your modelling steps. Your solution should include both your code (which markers must be able to run independently) and your rationale. This section must include:

- The fitting of a preliminary model on your preprocessed training data (to estimate some baseline performance level and to make sure your data is ready for modelling). Make this a simple model - logistic regression, decision trees or kNN are good options (but other may be used instead, if you prefer).
- The building of a predictive pipeline encapsulating all the learned/learnable aspects of your solution. This should result in a pipeline-type object that uses your raw training data (post EDA and test set splitting, but prior to any preprocessing), fits all your preprocessing transformations (incl. dimensionality reduction) and the predictive model using that data, and can then be used to ingest your test data (or any new data in the same format) and generate predictions for each entry.

Please refer to the CW specs for other requirements of this step (including, e.g., the addition of a dimensionality reduction step into your pipeline).

You can use as many code blocks as needed, but don't overdo it (remember - everything needs to have a proper rationale). If you compare different models, make sure to make it clear, and to indicate at the end which pipeline option is your final selected one.

3.1 Grouped cross-validation

The performance of the models are assessed using grouped 5-fold cross-validation with F1 score, accuracy, sensitivity, specificity and ROC curve as a metric.

3.2 Baseline model: Logistic regression

Before the finalisation of the model, with Logistic regression as the baseline model, PCA retaining 90% and 95% of the variance is compared. Besides showing the ideal number of PCs required for the model, these results also serve as a validation that the preprocessing recipe is working as intended.

```
## # A tibble: 5 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 accuracy binary    0.651     5 0.0194 pre0_mod0_post0
## 2 f_meas  binary    0.476     5 0.0114 pre0_mod0_post0
## 3 npv     binary    0.671     5 0.0284 pre0_mod0_post0
## 4 ppv     binary    0.592     5 0.0197 pre0_mod0_post0
## 5 roc_auc binary    0.669     5 0.00971 pre0_mod0_post0
```

```
## # A tibble: 5 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 accuracy binary    0.644     5 0.0190 pre0_mod0_post0
## 2 f_meas  binary    0.469     5 0.00955 pre0_mod0_post0
## 3 npv     binary    0.667     5 0.0289 pre0_mod0_post0
## 4 ppv     binary    0.578     5 0.0156 pre0_mod0_post0
## 5 roc_auc binary    0.663     5 0.00950 pre0_mod0_post0
```

Since grouped 5-fold Cross-validation show almost identical accuracy, F1 and ROC AUC for the 90% and 95% PCA thresholds, the lower-dimensional 90% model will be chosen for the final recipe.

3.3 Model comparison: logistic vs random forest

To assess whether predictive performance varies across different modelling approaches, a penalized logistic regression model and random forest model were compared against a baseline logistic regression model.

```
## Warning: No tuning parameters have been detected, performance will be evaluated using
## the resamples with no tuning.
## Did you want to assign any parameters with a value of 'tune()'?
```

```
## # A tibble: 5 x 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>      <dbl> <int>   <dbl> <chr>
## 1 accuracy binary    0.651     5 0.0194 pre0_mod0_post0
## 2 f_meas  binary    0.476     5 0.0114 pre0_mod0_post0
## 3 npv     binary    0.671     5 0.0284 pre0_mod0_post0
## 4 ppv     binary    0.592     5 0.0197 pre0_mod0_post0
## 5 roc_auc binary    0.669     5 0.00971 pre0_mod0_post0
```

```
## > A | warning: While computing binary 'precision()', no predicted events were detected (i.e.
##   'true_positive + false_positive = 0').
##   Precision is undefined in this case, and 'NA' will be returned.
##   Note that 467 true event(s) actually occurred for the problematic event level,
##   epitope
## There were issues with some computations A: x1There were issues with some computations A: x1
```

```
## # A tibble: 100 x 7
##   penalty .metric .estimator mean      n std_err .config
##   <dbl> <chr>    <chr>    <dbl> <int>    <dbl> <chr>
## 1 0.0001 accuracy binary    0.652     5 0.0190 pre0_mod01_post0
## 2 0.0001 f_meas  binary    0.470     5 0.0113 pre0_mod01_post0
## 3 0.0001 npv    binary    0.669     5 0.0279 pre0_mod01_post0
## 4 0.0001 ppv    binary    0.597     5 0.0207 pre0_mod01_post0
## 5 0.0001 roc_auc binary    0.669     5 0.00969 pre0_mod01_post0
## 6 0.000162 accuracy binary    0.652     5 0.0190 pre0_mod02_post0
## 7 0.000162 f_meas  binary    0.470     5 0.0113 pre0_mod02_post0
## 8 0.000162 npv    binary    0.669     5 0.0279 pre0_mod02_post0
## 9 0.000162 ppv    binary    0.597     5 0.0207 pre0_mod02_post0
## 10 0.000162 roc_auc binary    0.669     5 0.00969 pre0_mod02_post0
## # i 90 more rows
```

```
## # A tibble: 20 x 8
##   mtry min_n .metric .estimator mean      n std_err .config
##   <int> <int> <chr>    <chr>    <dbl> <int>    <dbl> <chr>
## 1 10 5 accuracy binary    0.704     5 0.0161 pre0_mod1_post0
## 2 10 5 f_meas  binary    0.578     5 0.00758 pre0_mod1_post0
## 3 10 5 npv    binary    0.717     5 0.0268 pre0_mod1_post0
## 4 10 5 ppv    binary    0.668     5 0.0170 pre0_mod1_post0
## 5 10 5 roc_auc binary    0.766     5 0.0144 pre0_mod1_post0
## 6 10 20 accuracy binary    0.698     5 0.0197 pre0_mod2_post0
## 7 10 20 f_meas  binary    0.576     5 0.00765 pre0_mod2_post0
## 8 10 20 npv    binary    0.715     5 0.0300 pre0_mod2_post0
## 9 10 20 ppv    binary    0.656     5 0.00716 pre0_mod2_post0
## 10 10 20 roc_auc binary    0.766     5 0.0153 pre0_mod2_post0
## 11 60 5 accuracy binary    0.697     5 0.0152 pre0_mod3_post0
## 12 60 5 f_meas  binary    0.588     5 0.00649 pre0_mod3_post0
## 13 60 5 npv    binary    0.723     5 0.0267 pre0_mod3_post0
## 14 60 5 ppv    binary    0.642     5 0.0183 pre0_mod3_post0
## 15 60 5 roc_auc binary    0.763     5 0.0143 pre0_mod3_post0
## 16 60 20 accuracy binary    0.698     5 0.0184 pre0_mod4_post0
## 17 60 20 f_meas  binary    0.593     5 0.00976 pre0_mod4_post0
## 18 60 20 npv    binary    0.726     5 0.0294 pre0_mod4_post0
## 19 60 20 ppv    binary    0.640     5 0.0164 pre0_mod4_post0
## 20 60 20 roc_auc binary    0.760     5 0.0153 pre0_mod4_post0
```

```
## # A tibble: 3 x 10
##   .metric .estimator mean      n std_err .config      model penalty mtry min_n
##   <chr>    <chr>    <dbl> <int>    <dbl> <chr>    <chr>    <dbl> <int> <int>
## 1 f_meas  binary    0.476     5 0.0114 pre0_mod0_po~ logi~ NA      NA      NA
## 2 f_meas  binary    0.470     5 0.0113 pre0_mod01_p~ pena~ 0.0001 NA      NA
## 3 f_meas  binary    0.593     5 0.00976 pre0_mod4_po~ rand~ NA      60     20
```

The comparison of F1-score between the two models show that the random forest model with F1 score ≈ 0.60 has better score than logistic regression model with ≈ 0.48 and penalized logistic regression model with ≈ 0.47 . Therefore, the random forest model is selected for further evaluation.

4. Model assessment

Note: It is expected that this section should result in no more than 100-ish words (possibly less; not counting the code blocks).

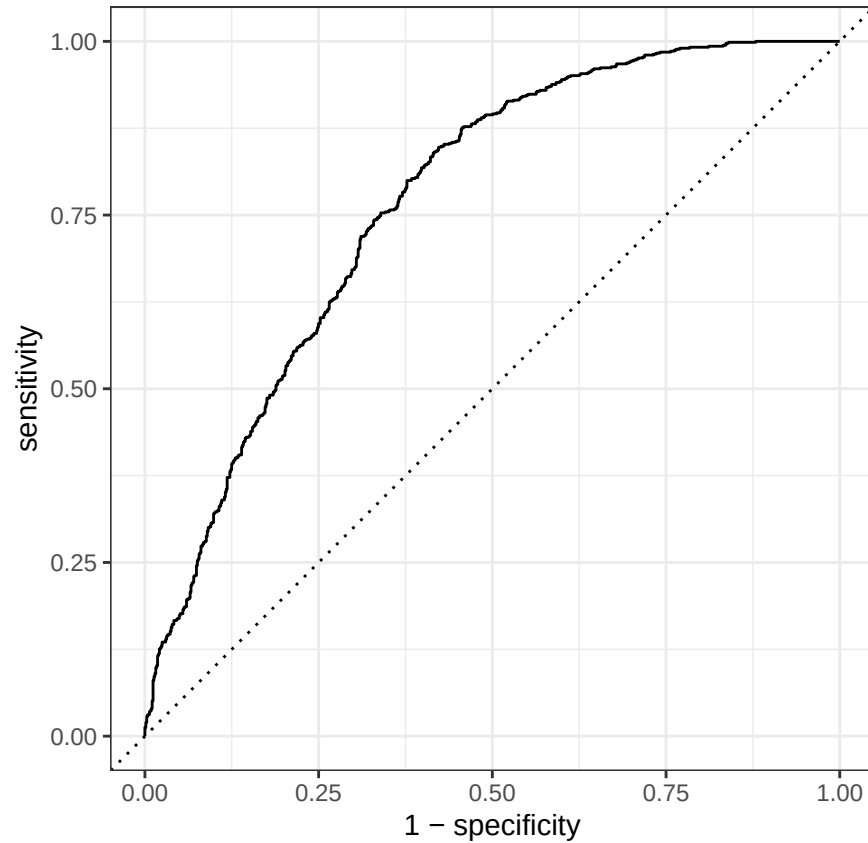
This is a short section in which you must estimate the generalisation performance of your model on the test set that you isolated at the start of your Data preprocessing step. It is also a chance for you to ensure that your pipeline object can take in new data and return the required predictions.

Use your pipeline to generate predictions for your test set, and calculate the observed performance using the F_1 -score as your main performance indicator. You can also additionally report other performance metrics if you think it's relevant.

You can use as many code blocks as needed, but it's likely that this section can be completed with a single one. Make sure that your performance indicators are clearly echoed, so that a reader can easily check if after re-running your script. Add a short comment on the observed performance (is it good? Is it poor? Remember that you're not being assessed on the performance of your models, but on the soundness of your approach).

Hyperparameter tuned random forest model is refitted on the whole training set with finalized preprocessing recipe. It is then used to generate predictions for the test data which will provide an unbiased estimate of how well the model is expected to perform on new, unseen data such as the held-out test set.

```
## # A tibble: 5 x 4
##   .metric .estimator .estimate .config
##   <chr>    <chr>         <dbl> <chr>
## 1 accuracy binary          0.673 pre0_mod0_post0
## 2 ppv      binary          0.705 pre0_mod0_post0
## 3 npv      binary          0.653 pre0_mod0_post0
## 4 f_meas   binary          0.615 pre0_mod0_post0
## 5 roc_auc  binary          0.768 pre0_mod0_post0
```



```
## # A tibble: 5 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>      <dbl>
## 1 accuracy binary      0.664
## 2 ppv     binary      0.693
## 3 npv     binary      0.648
## 4 f_meas  binary      0.607
## 5 roc_auc binary      0.762
```

```
##           Truth
## Prediction  epitope non-epitope
##   epitope      383      170
##   non-epitope  326      599
```

The final random forest model achieved an accuracy of 0.66, F1-score of ≈ 0.60 , precision(PPV) of ≈ 0.69 and Negative predictive value(NPV) of ≈ 0.65 , with a ROC AUC of ≈ 0.76 . The confusion matrix shows that the model correctly identifies about half of the true epitopes (true positives vs false negatives) and reasonably often rejects non-epitopes (true negatives vs false positives), indicating moderate rather than outstanding predictive performance. This moderate performance may be due to difficulty of the epitope prediction problem, the limited discriminative power of the available features, and the relatively simple tuning of the random forest hyperparameters. Overall, these results suggest that the model captures some useful signal for epitope prediction but still leaves substantial room for improvement.

5. Model deployment

Note: This section is not expected to have much text in it, only the code and the results.

Finally, in this section you'll generate predictions for some data for which you do not know the class labels. This simulates the real-life scenario in which your model needs to be deployed to generate new predictions for new, unknown data (after all, if the labels were known, we wouldn't need a model).

Make sure that the data file **df_holdout.rds** is placed in the same folder as this Task02.Rmd file. The new data will then be automatically loaded and set up for you, and stored into a dataframe variable called **df_holdout**, with the same columns as the original **df** that was loaded at the start of this task.

- Use your fitted pipeline to generate predictions for this new data.
- Build a new data frame called containing the following columns:
 - **Info_PepID** (taken from **df_holdout**)
 - **Info_pos** (taken from **df_holdout**)
 - **Predicted_class** (with the predictions produced by your pipeline)
- Save your predictions dataframe to a file called **mypreds.rds**. You will be able to submit this file, and it will be used to give you some feedback on the actual performance of your pipeline on new data.

The last step is to deploy the final trained model to the unseen holdout data ("**df_holdout**") to generate predictions, which are then saved for external evaluation.

```
## # A tibble: 6 x 389
##   Info_PepID Info_group Info_protein_id Info_pos feat_local_Entropy
##   <chr>      <int> <chr>          <dbl>      <dbl>
## 1 0704243A:10      1 0704243A      222.      3.22
## 2 0704243A:12      1 0704243A      303.      3.28
## 3 0704243A:14      1 0704243A      233.      3.21
## 4 0704243A:16      1 0704243A      276.      3.08
## 5 0704243A:18      1 0704243A      316.      3.34
## 6 0704243A:2       1 0704243A      179.      3.35
## # i 384 more variables: feat_local_MolWeight <dbl>,
## #   feat_local_AAtypes_Tiny <dbl>, feat_local_AAtypes_Small <dbl>,
## #   feat_local_AAtypes_Aliphatic <dbl>, feat_local_AAtypes_Aromatic <dbl>,
## #   feat_local_AAtypes_NonPolar <dbl>, feat_local_AAtypes_Polar <dbl>,
## #   feat_local_AAtypes_Charged <dbl>, feat_local_AAtypes_Basic <dbl>,
## #   feat_local_AAtypes_Acidic <dbl>, feat_local_Atoms_nC <dbl>,
## #   feat_local_Atoms_nH <dbl>, feat_local_Atoms_nN <dbl>, ...

## Rows: 3,177
## Columns: 2
## $ Info_ID      <chr> "0704243A:10_221.9375", "0704243A:12_303.072727272727", "07~
## $ Info_group   <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 10, 10, 10, 10, 10, 10, 10, 202,~

## Warning: Novel levels found in column "Info_group": "1", "10", "11", "21", "38", "166",
## "202", "224", "356", and "369".
## i The levels have been removed, and values have been coerced to <NA>.
## # A tibble: 3,177 x 3
##   Info_PepID Info_pos Predicted_class
##   <chr>      <dbl> <fct>
## 1 0704243A:10 221.9375 1
## 2 0704243A:12 303.0727 1
## 3 0704243A:14 233.0000 1
## 4 0704243A:16 276.0000 1
## 5 0704243A:18 316.0000 1
## 6 0704243A:2  179.0000 1
## 7 0704243A:1  100.0000 1
## 8 0704243A:3  150.0000 1
## 9 0704243A:4  200.0000 1
## 10 0704243A:5  250.0000 1
## 11 0704243A:6  300.0000 1
## 12 0704243A:7  350.0000 1
## 13 0704243A:8  400.0000 1
## 14 0704243A:9  450.0000 1
## 15 0704243A:10 500.0000 1
## 16 0704243A:11 550.0000 1
## 17 0704243A:12 600.0000 1
## 18 0704243A:13 650.0000 1
## 19 0704243A:14 700.0000 1
## 20 0704243A:15 750.0000 1
## 21 0704243A:16 800.0000 1
## 22 0704243A:17 850.0000 1
## 23 0704243A:18 900.0000 1
## 24 0704243A:19 950.0000 1
## 25 0704243A:20 1000.0000 1
## 26 0704243A:21 1050.0000 1
## 27 0704243A:22 1100.0000 1
## 28 0704243A:23 1150.0000 1
## 29 0704243A:24 1200.0000 1
## 30 0704243A:25 1250.0000 1
## 31 0704243A:26 1300.0000 1
## 32 0704243A:27 1350.0000 1
## 33 0704243A:28 1400.0000 1
## 34 0704243A:29 1450.0000 1
## 35 0704243A:30 1500.0000 1
## 36 0704243A:31 1550.0000 1
## 37 0704243A:32 1600.0000 1
## 38 0704243A:33 1650.0000 1
## 39 0704243A:34 1700.0000 1
## 40 0704243A:35 1750.0000 1
## 41 0704243A:36 1800.0000 1
## 42 0704243A:37 1850.0000 1
## 43 0704243A:38 1900.0000 1
## 44 0704243A:39 1950.0000 1
## 45 0704243A:40 2000.0000 1
## 46 0704243A:41 2050.0000 1
## 47 0704243A:42 2100.0000 1
## 48 0704243A:43 2150.0000 1
## 49 0704243A:44 2200.0000 1
## 50 0704243A:45 2250.0000 1
## 51 0704243A:46 2300.0000 1
## 52 0704243A:47 2350.0000 1
## 53 0704243A:48 2400.0000 1
## 54 0704243A:49 2450.0000 1
## 55 0704243A:50 2500.0000 1
## 56 0704243A:51 2550.0000 1
## 57 0704243A:52 2600.0000 1
## 58 0704243A:53 2650.0000 1
## 59 0704243A:54 2700.0000 1
## 60 0704243A:55 2750.0000 1
## 61 0704243A:56 2800.0000 1
## 62 0704243A:57 2850.0000 1
## 63 0704243A:58 2900.0000 1
## 64 0704243A:59 2950.0000 1
## 65 0704243A:60 3000.0000 1
## 66 0704243A:61 3050.0000 1
## 67 0704243A:62 3100.0000 1
## 68 0704243A:63 3150.0000 1
## 69 0704243A:64 3200.0000 1
## 70 0704243A:65 3250.0000 1
## 71 0704243A:66 3300.0000 1
## 72 0704243A:67 3350.0000 1
## 73 0704243A:68 3400.0000 1
## 74 0704243A:69 3450.0000 1
## 75 0704243A:70 3500.0000 1
## 76 0704243A:71 3550.0000 1
## 77 0704243A:72 3600.0000 1
## 78 0704243A:73 3650.0000 1
## 79 0704243A:74 3700.0000 1
## 80 0704243A:75 3750.0000 1
## 81 0704243A:76 3800.0000 1
## 82 0704243A:77 3850.0000 1
## 83 0704243A:78 3900.0000 1
## 84 0704243A:79 3950.0000 1
## 85 0704243A:80 4000.0000 1
## 86 0704243A:81 4050.0000 1
## 87 0704243A:82 4100.0000 1
## 88 0704243A:83 4150.0000 1
## 89 0704243A:84 4200.0000 1
## 90 0704243A:85 4250.0000 1
## 91 0704243A:86 4300.0000 1
## 92 0704243A:87 4350.0000 1
## 93 0704243A:88 4400.0000 1
## 94 0704243A:89 4450.0000 1
## 95 0704243A:90 4500.0000 1
## 96 0704243A:91 4550.0000 1
## 97 0704243A:92 4600.0000 1
## 98 0704243A:93 4650.0000 1
## 99 0704243A:94 4700.0000 1
## 100 0704243A:95 4750.0000 1
## 101 0704243A:96 4800.0000 1
## 102 0704243A:97 4850.0000 1
## 103 0704243A:98 4900.0000 1
## 104 0704243A:99 4950.0000 1
## 105 0704243A:100 5000.0000 1
## 106 0704243A:101 5050.0000 1
## 107 0704243A:102 5100.0000 1
## 108 0704243A:103 5150.0000 1
## 109 0704243A:104 5200.0000 1
## 110 0704243A:105 5250.0000 1
## 111 0704243A:106 5300.0000 1
## 112 0704243A:107 5350.0000 1
## 113 0704243A:108 5400.0000 1
## 114 0704243A:109 5450.0000 1
## 115 0704243A:110 5500.0000 1
## 116 0704243A:111 5550.0000 1
## 117 0704243A:112 5600.0000 1
## 118 0704243A:113 5650.0000 1
## 119 0704243A:114 5700.0000 1
## 120 0704243A:115 5750.0000 1
## 121 0704243A:116 5800.0000 1
## 122 0704243A:117 5850.0000 1
## 123 0704243A:118 5900.0000 1
## 124 0704243A:119 5950.0000 1
## 125 0704243A:120 6000.0000 1
## 126 0704243A:121 6050.0000 1
## 127 0704243A:122 6100.0000 1
## 128 0704243A:123 6150.0000 1
## 129 0704243A:124 6200.0000 1
## 130 0704243A:125 6250.0000 1
## 131 0704243A:126 6300.0000 1
## 132 0704243A:127 6350.0000 1
## 133 0704243A:128 6400.0000 1
## 134 0704243A:129 6450.0000 1
## 135 0704243A:130 6500.0000 1
## 136 0704243A:131 6550.0000 1
## 137 0704243A:132 6600.0000 1
## 138 0704243A:133 6650.0000 1
## 139 0704243A:134 6700.0000 1
## 140 0704243A:135 6750.0000 1
## 141 0704243A:136 6800.0000 1
## 142 0704243A:137 6850.0000 1
## 143 0704243A:138 6900.0000 1
## 144 0704243A:139 6950.0000 1
## 145 0704243A:140 7000.0000 1
## 146 0704243A:141 7050.0000 1
## 147 0704243A:142 7100.0000 1
## 148 0704243A:143 7150.0000 1
## 149 0704243A:144 7200.0000 1
## 150 0704243A:145 7250.0000 1
## 151 0704243A:146 7300.0000 1
## 152 0704243A:147 7350.0000 1
## 153 0704243A:148 7400.0000 1
## 154 0704243A:149 7450.0000 1
## 155 0704243A:150 7500.0000 1
## 156 0704243A:151 7550.0000 1
## 157 0704243A:152 7600.0000 1
## 158 0704243A:153 7650.0000 1
## 159 0704243A:154 7700.0000 1
## 160 0704243A:155 7750.0000 1
## 161 0704243A:156 7800.0000 1
## 162 0704243A:157 7850.0000 1
## 163 0704243A:158 7900.0000 1
## 164 0704243A:159 7950.0000 1
## 165 0704243A:160 8000.0000 1
## 166 0704243A:161 8050.0000 1
## 167 0704243A:162 8100.0000 1
## 168 0704243A:163 8150.0000 1
## 169 0704243A:164 8200.0000 1
## 170 0704243A:165 8250.0000 1
## 171 0704243A:166 8300.0000 1
## 172 0704243A:167 8350.0000 1
## 173 0704243A:168 8400.0000 1
## 174 0704243A:169 8450.0000 1
## 175 0704243A:170 8500.0000 1
## 176 0704243A:171 8550.0000 1
## 177 0704243A:172 8600.0000 1
## 178 0704243A:173 8650.0000 1
## 179 0704243A:174 8700.0000 1
## 180 0704243A:175 8750.0000 1
## 181 0704243A:176 8800.0000 1
## 182 0704243A:177 8850.0000 1
## 183 0704243A:178 8900.0000 1
## 184 0704243A:179 8950.0000 1
## 185 0704243A:180 9000.0000 1
## 186 0704243A:181 9050.0000 1
## 187 0704243A:182 9100.0000 1
## 188 0704243A:183 9150.0000 1
## 189 0704243A:184 9200.0000 1
## 190 0704243A:185 9250.0000 1
## 191 0704243A:186 9300.0000 1
## 192 0704243A:187 9350.0000 1
## 193 0704243A:188 9400.0000 1
## 194 0704243A:189 9450.0000 1
## 195 0704243A:190 9500.0000 1
## 196 0704243A:191 9550.0000 1
## 197 0704243A:192 9600.0000 1
## 198 0704243A:193 9650.0000 1
## 199 0704243A:194 9700.0000 1
## 200 0704243A:195 9750.0000 1
## 201 0704243A:196 9800.0000 1
## 202 0704243A:197 9850.0000 1
## 203 0704243A:198 9900.0000 1
## 204 0704243A:199 9950.0000 1
## 205 0704243A:200 10000.0000 1
## 206 0704243A:201 10050.0000 1
## 207 0704243A:202 10100.0000 1
## 208 0704243A:203 10150.0000 1
## 209 0704243A:204 10200.0000 1
## 210 0704243A:205 10250.0000 1
## 211 0704243A:206 10300.0000 1
## 212 0704243A:207 10350.0000 1
## 213 0704243A:208 10400.0000 1
## 214 0704243A:209 10450.0000 1
## 215 0704243A:210 10500.0000 1
## 216 0704243A:211 10550.0000 1
## 217 0704243A:212 10600.0000 1
## 218 0704243A:213 10650.0000 1
## 219 0704243A:214 10700.0000 1
## 220 0704243A:215 10750.0000 1
## 221 0704243A:216 10800.0000 1
## 222 0704243A:217 10850.0000 1
## 223 0704243A:218 10900.0000 1
## 224 0704243A:219 10950.0000 1
## 225 0704243A:220 11000.0000 1
## 226 0704243A:221 11050.0000 1
## 227 0704243A:222 11100.0000 1
## 228 0704243A:223 11150.0000 1
## 229 0704243A:224 11200.0000 1
## 230 0704243A:225 11250.0000 1
## 231 0704243A:226 11300.0000 1
## 232 0704243A:227 11350.0000 1
## 233 0704243A:228 11400.0000 1
## 234 0704243A:229 11450.0000 1
## 235 0704243A:230 11500.0000 1
## 236 0704243A:231 11550.0000 1
## 237 0704243A:232 11600.0000 1
## 238 0704243A:233 11650.0000 1
## 239 0704243A:234 11700.0000 1
## 240 0704243A:235 11750.0000 1
## 241 0704243A:236 11800.0000 1
## 242 0704243A:237 11850.0000 1
## 243 0704243A:238 11900.0000 1
## 244 0704243A:239 11950.0000 1
## 245 0704243A:240 12000.0000 1
## 246 0704243A:241 12050.0000 1
## 247 0704243A:242 12100.0000 1
## 248 0704243A:243 12150.0000 1
## 249 0704243A:244 12200.0000 1
## 250 0704243A:245 12250.0000 1
## 251 0704243A:246 12300.0000 1
## 252 0704243A:247 12350.0000 1
## 253 0704243A:248 12400.0000 1
## 254 0704243A:249 12450.0000 1
## 255 0704243A:250 12500.0000 1
## 256 0704243A:251 12550.0000 1
## 257 0704243A:252 12600.0000 1
## 258 0704243A:253 12650.0000 1
## 259 0704243A:254 12700.0000 1
## 260 0704243A:255 12750.0000 1
## 261 0704243A:256 12800.0000 1
## 262 0704243A:257 12850.0000 1
## 263 0704243A:258 12900.0000 1
## 264 0704243A:259 12950.0000 1
## 265 0704243A:260 13000.0000 1
## 266 0704243A:261 13050.0000 1
## 267 0704243A:262 13100.0000 1
## 268 0704243A:263 13150.0000 1
## 269 0704243A:264 13200.0000 1
## 270 0704243A:265 13250.0000 1
## 271 0704243A:266 13300.0000 1
## 272 0704243A:267 13350.0000 1
## 273 0704243A:268 13400.0000 1
## 274 0704243A:269 13450.0000 1
## 275 0704243A:270 13500.0000 1
## 276 0704243A:271 13550.0000 1
## 277 0704243A:272 13600.0000 1
## 278 0704243A:273 13650.0000 1
## 279 0704243A:274 13700.0000 1
## 280 0704243A:275 13750.0000 1
## 281 0704243A:276 13800.0000 1
## 282 0704243A:277 13850.0000 1
## 283 0704243A:278 13900.0000 1
## 284 0704243A:279 13950.0000 1
## 285 0704243A:280 14000.0000 1
## 286 0704243A:281 14050.0000 1
## 287 0704243A:282 14100.0000 1
## 288 0704243A:283 14150.0000 1
## 289 0704243A:284 14200.0000 1
## 290 0704243A:285 14250.0000 1
## 291 0704243A:286 14300.0000 1
## 292 0704243A:287 14350.0000 1
## 293 0704243A:288 14400.0000 1
## 294 0704243A:289 14450.0000 1
## 295 0704243A:290 14500.0000 1
## 296 0704243A:291 14550.0000 1
## 297 0704243A:292 14600.0000 1
## 298 0704243A:293 14650.0000 1
## 299 0704243A:294 14700.0000 1
## 300 0704243A:295 14750.0000 1
## 301 0704243A:296 14800.0000 1
## 302 0704243A:297 14850.0000 1
## 303 0704243A:298 14900.0000 1
## 304 0704243A:299 14950.0000 1
## 305 0704243A:300 15000.0000 1
## 306 0704243A:301 15050.0000 1
## 307 0704243A:302 15100.0000 1
## 308 0704243A:303 15150.0000 1
## 309 0704243A:304 15200.0000 1
## 310 0704243A:305 15250.0000 1
## 311 0704243A:306 15300.0000 1
## 312 0704243A:307 15350.0000 1
## 313 0704243A:308 15400.0000 1
## 314 0704243A:309 15450.0000 1
## 315 0704243A:310 15500.0000 1
## 316 0704243A:311 15550.0000 1
## 317 0704243A:312 15600.0000 1
## 318 0704243A:313 15650.0000 1
## 319 0704243A:314 15700.0000 1
## 320 0704243A:315 15750.0000 1
## 321 0704243A:316 15800.0000 1
## 322 0704243A:317 15850.0000 1
## 323 0704243A:318 15900.0000 1
## 324 0704243A:319 15950.0000 1
## 325 0704243A:320 16000.0000 1
## 326 0704243A:321 16050.0000 1
## 327 0704243A:322 16100.0000 1
## 328 0704243A:323 16150.0000 1
## 329 0704243A:324 16200.0000 1
## 330 0704243A:325 16250.0000 1
## 331 0704243A:326 16300.0000 1
## 332 0704243A:327 16350.0000 1
## 333 0704243A:328 16400.0000 1
## 334 0704243A:329 16450.0000 1
## 335 0704243A:330 16500.0000 1
## 336 0704243A:331 16550.0000 1
## 337 0704243A:332 16600.0000 1
## 338 0704243A:333 16650.0000 1
## 339 0704243A:334 16700.0000 1
## 340 0704243A:335 16750.0000 1
## 341 0704243A:336 16800.0000 1
## 342 0704243A:337 16850.0000 1
## 343 0704243A:338 16900.0000 1
## 344 0704243A:339 16950.0000 1
## 345 0704243A:340 17000.0000 1
## 346 0704243A:341 17050.0000 1
## 347 0704243A:342 17100.0000 1
## 348 0704243A:343 17150.0000 1
## 349 0704243A:344 17200.0000 1
## 350 0704243A:345 17250.0000 1
## 351 0704243A:346 17300.0000 1
## 352 0704243A:347 17350.0000 1
## 353 0704243A:348 17400.0000 1
## 354 0704243A:349 17450.0000 1
## 355 0704243A:350 17500.0000 1
## 356 0704243A:351 17550.0000 1
## 357 0704243A:352 17600.0000
```

```
## 1 0704243A:10 222. non-epitope
## 2 0704243A:12 303. epitope
## 3 0704243A:14 233 non-epitope
## 4 0704243A:16 276 non-epitope
## 5 0704243A:18 316. epitope
## 6 0704243A:2 179. non-epitope
## 7 0704243A:4 131 non-epitope
## 8 0704243A:6 185. non-epitope
## 9 0704243A:8 182. non-epitope
## 10 0704243B:1 6.70 epitope
## # i 3,167 more rows
```

===== End of Task II =====